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Review article

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Redefining the central pattern generator for vertebrate locomotion

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Abstract

Since its original formulation, the concept of the central pattern generator (CPG) has provided a foundational framework for understanding vertebrate locomotion. Recent advances in circuit-level neuroscience in zebrafish have redefined the CPG as a dynamic, modular and hybrid sensorimotor system. A central shift has been the replacement of the classical view of the CPG as a unitary rhythm generator with the idea that it is made up of speed-specific modules that are recruited via gear-shifting mechanisms tailored to behavioural demands. Brainstem circuits have emerged as layered controllers that initiate locomotion and modulate episode duration, speed and direction, whereas motor neurons and proprioceptors are now recognized as integral CPG components that shape rhythm and coordination. Here, I use zebrafish as a primary reference point — alongside explicit comparisons to conserved and divergent principles in mammals — to highlight new facets of the vertebrate CPG that redefine it as a highly adaptable, multilayered control system that is continuously tuned by sensory feedback and descending input. This offers a roadmap for decoding the neural logic of adaptive movement across contexts and evolutionary scales and highlights how principles revealed in zebrafish can provide testable hypotheses for terrestrial vertebrates.

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Introduction

Locomotion is essential for life, enabling organisms to explore, interact with and adapt to their environments[1–3]. Across the animal kingdom, the forms of locomotion are as diverse as the species that exhibit them, from the precise flight of birds to the undulating swimming of fish and the complex gaits of mammals. Despite this diversity, the basic neural principles governing locomotion are remarkably conserved across species[4–19]. Studies of how the nervous system enables organisms to move have provided critical insights into how the nervous system coordinates behaviour, offering a powerful lens that allows us to understand how the brain and spinal cord integrate sensory inputs, internal motor programmes and biomechanical constraints to generate purposeful behaviour.

Locomotion is inherently rhythmic, consisting of repeating cycles of muscle activity. In limbed animals, these cycles are typically divided into a stance phase (providing propulsion and stability) and a swing phase (repositioning the limb). Coordinated movement requires precise timing between these phases, as well as alternation between the left and right sides of the body and reciprocal activation of flexor and extensor muscles around joints. As discussed in this Review, the central pattern generator (CPG) is the core spinal machinery responsible for generating these rhythms and coordination patterns.

One hallmark of locomotion is its flexibility — speed, direction and gait are rapidly adjusted to meet changing goals or environmental conditions. In vertebrates, this flexibility arises from dynamic interactions between supraspinal centres, which select context-appropriate motor programmes, and spinal circuits, which implement them through coordinated muscle activation[4,12,20–27] (Fig. 1). Together, these systems ensure that locomotor movement parameters remain aligned with the behavioural context. Low-speed locomotion often supports exploration or foraging, whereas high-speed movement enables rapid escape or pursuit. Directional changes allow navigation towards goals or away from threats, and in quadrupeds, increasing speed is coupled to discrete gait transitions, from walking to trotting to galloping.

Conceptual advances in understanding how locomotor movements are initiated, generated and modulated have been shaped by both comparative approaches and technological innovation[6,12–14,20,23,24,28–31]. Early debates centred on whether the rhythmic movements that underlie locomotion arise from reflex chains wherein sensory feedback from muscle contraction during one part of the step cycle automatically activates the set of muscles needed for the subsequent phase in a chain, or from intrinsic spinal networks. These debates culminated in the classical CPG model in which a self-contained spinal circuit is capable

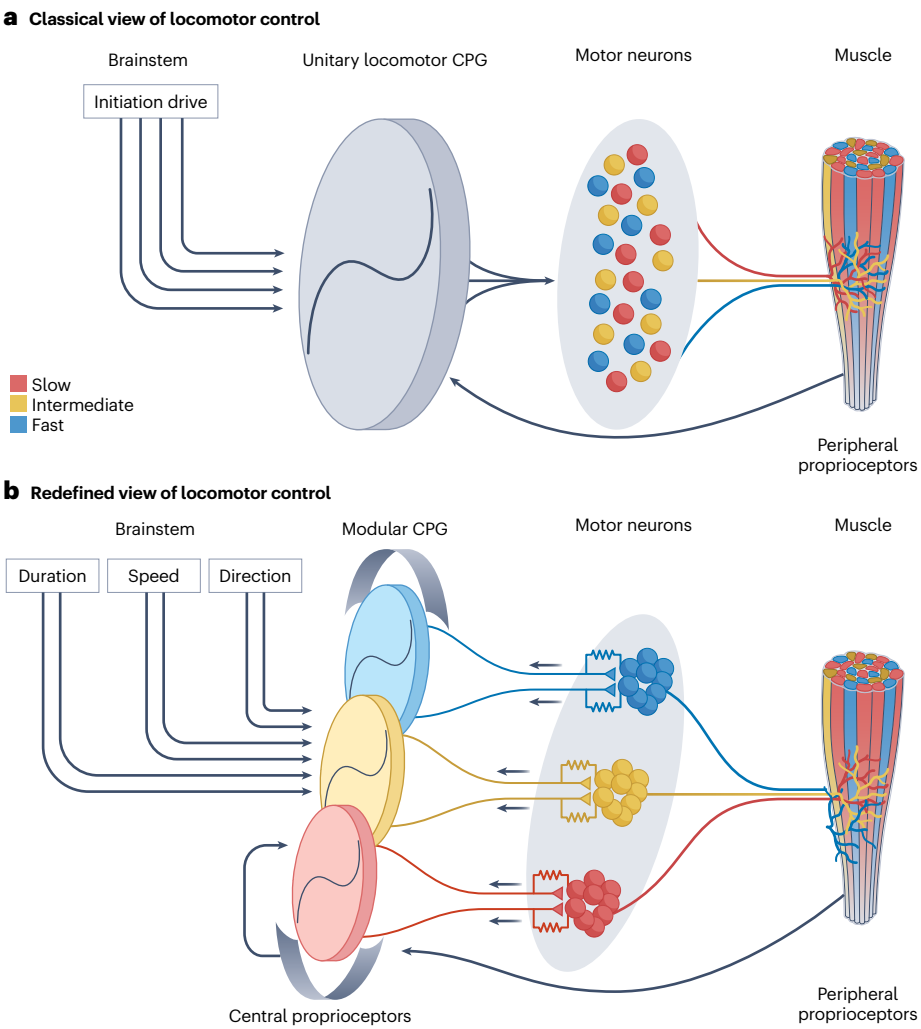


Fig. 1 | Classical versus redefined views of locomotor control. a, According to classical views of locomotor control, locomotor movements are generated by a single, unitary central pattern generator (CPG) consisting of an interconnected network of interneurons. This CPG produces a common rhythmic output that is broadcast to all motor neuron subtypes (slow (red), intermediate (yellow) and fast (blue)), which innervate the corresponding muscle fibre types. Motor neurons are viewed primarily as passive relays, faithfully transmitting CPG activity to the muscles. Locomotion is initiated when descending commands from the brainstem provide a general 'go' signal that activates the CPG. In this framework, locomotor speed is determined by increased excitatory drive from the brainstem, which sequentially recruits motor neuron subtypes (slow, then intermediate, and finally fast), thereby scaling muscle output to the behavioural demands. Sensory feedback is viewed as arising exclusively from proprioceptors located in the periphery. **b**, The redefined view proposes that the locomotor CPG is organized into three speed-specific circuit modules that are recruited sequentially with increased speed. At slow speeds, only the slow module is active, selectively activating its slow motor neurons and their corresponding muscle fibres. Increasing speed recruits the intermediate module and ultimately the fast module, activating their respective motor neurons. In this framework, motor neurons are active circuit components that exert a retrograde influence on CPG interneurons through gap junctions. Furthermore, the existence of centrally located proprioceptors provides short-latency sensory feedback that allows these neurons to directly influence the timing of CPG activity.

of generating rhythmic output independently of sensory input[5,8,10,12,13] (Fig. 1a). However, the subsequent advent of molecular, genetic, optical and electrophysiological tools revealed a more complex picture. In particular, the application of these tools in zebrafish – a model that affords unparalleled access to molecularly identified neurons and enables complete circuit-level analysis – has revealed organizational principles that complement and extend findings from mammalian models, whose CPG organization remains elusive. Studies in adult zebrafish have shown that locomotor CPG circuits are modular, that motor neurons and proprioceptors are integrated as core elements, and that brainstem circuits exert layered control over the initiation, speed and direction of locomotion (Fig. 1b). These discoveries, thus, challenge the notion of a unitary rhythm generator and call for a conceptual shift towards a distributed, modular and hybrid sensorimotor CPG.

In this Review, I synthesize the recent insights into the brainstem and spinal circuits underlying vertebrate locomotion. I place an emphasis on discoveries from zebrafish, because these serve as a powerful reference point for understanding fundamental principles that are conserved across vertebrates. Comparisons with mammalian systems, particularly the mouse, highlight shared circuit motifs and evolutionary adaptations. I emphasize the principles of modular organization, the role of molecular identity in circuit assembly and function, and the dynamic interplay between intrinsic spinal activity, sensory feedback and descending control. In this framework, spinal CPG modules are flexibly recruited by brainstem commands, continuously tuned by proprioceptive input and repurposed across behavioural contexts. This positions vertebrate locomotor circuits as reusable building blocks whose flexible deployment underlies the adaptability of locomotor behaviour and provides a roadmap for decoding CPG logic across contexts and species.

Controllers of the CPG: brainstem circuits

Locomotion is driven by internal motivational states and directed towards specific goals that serve physiological or environmental needs. However, in moments of imminent threat, motor priorities shift dramatically and locomotion becomes reactive, prioritizing escape or shelter-seeking responses that may lack a clear end point. These flexible motor outputs are orchestrated by a distributed network of brain regions that converge on spinal projection neurons in the brainstem. These brainstem neurons encode the key parameters of locomotor behaviour: when to start, how fast to go, and in which direction to steer.

Initiation commands

Classical stimulation studies were the first to identify key locomotor control centres within the diencephalon and mesencephalon of a range of vertebrate species[12,20,23–25,32–38]. Among these, the mesencephalic locomotor region (MLR), which projects to reticulospinal neurons in the medulla, has emerged as a central hub for initiating forward locomotion and modulating speed[12,20,23,32,34,37,39–41]. In quadrupeds, graded stimulation of the MLR evokes locomotion with increases in speed that are associated with sequential transitions across gaits, linking MLR output to speed-dependent motor unit recruitment[12,20–22,24,25,32–34,38]. A parallel circuit arises from the diencephalic locomotor region (DLR), which can also initiate and sustain locomotor bouts, even following MLR lesions[12]. The DLR and MLR, thus, represent two distinct initiation systems that converge onto reticulospinal neurons to flexibly drive locomotion depending on behavioural state. Importantly, neither the MLR nor the DLR projects directly to the spinal cord, wherein the locomotor CPG resides. Instead, their commands are relayed through downstream brainstem circuits, highlighting the essential role of the brainstem in translating motivational drive into patterned locomotion[21,42]. In mice, a small subset of glutamatergic MLR neurons do project to the spinal cord, but these evoke body extension rather than locomotion[20,40].

Circuits setting boundaries and speed

Both duration and speed of locomotion must be finely regulated to match behavioural context. Recent studies in zebrafish and mice have revealed that these two parameters, how long to move for and how fast to go, are encoded by distinct but complementary brainstem circuits (Fig. 2a). In zebrafish, the nucleus of the medial longitudinal fasciculus (nMLF) in the midbrain contributes alongside the MLR to initiate locomotion and to control its speed and duration (as well as direction, see 'Defining direction and trajectory')[26,43–45], and the two structures interact rather than represent redundant systems (Fig. 2b). Although the mammalian homologue of the nMLF remains debated, elements of its function appear to be distributed across several midbrain nuclei, including the rostral interstitial nMLF and the interstitial nucleus of Cajal[20,40].

The zebrafish nMLF contains two subpopulations of glutamatergic neurons that project directly to the spinal cord and provide collateral input to hindbrain reticulospinal neurons[26,43,44]. Medially located vesicular glutamate transporter 2 (Vglut2)-expressing neurons encode bout onset and duration, setting the temporal boundaries of locomotor episodes[43]. Laterally positioned Vglut1-expressing neurons regulate transitions to high-speed swimming[43]. Selective ablation of these populations disrupts swim initiation and fast locomotor output, respectively, demonstrating that the timing and speed of locomotion are encoded largely by separate circuits[43,46] (Fig. 2a,b).

A similar logic tends to apply in mammals; however, in this case, it has been proposed that distinct circuits regulate episode duration and speed during either slow exploratory or high-speed locomotion. In mice, the MLR comprises the cuneiform nucleus (CnF), pedunculopontine nucleus (PPN) and mesencephalic reticular formation (mRT), each of which contains heterogeneous neuronal populations that are primarily defined by their neurotransmitter identity and projection targets[12,20,23] (Fig. 2c). Optogenetic studies in mice have revealed a speed-dependent functional specialization within the glutamatergic populations of the MLR. CnF glutamatergic neurons are robustly engaged during high-speed locomotion, particularly during escape-like responses[32,35,37,47]. These neurons project to the lateral paragigantocellular nucleus (LPGi) in the medulla, a brainstem relay that transmits MLR commands to spinal circuits[32,33,35,37,40,41] (Fig. 2c). By contrast, studies investigating the effects of activation of PPN glutamatergic neurons, which also project to LPGi, have produced variable results. Some studies support a role for these neurons in slow, exploratory locomotion[32] (Fig. 2c), whereas others report a suppression of ongoing movement following PPN stimulation[37,47]. Thus, the contribution of PPN glutamatergic neurons to locomotion remains unresolved, with the disparate findings probably reflecting anatomical heterogeneity, state dependence and incomplete resolution of functional subpopulations. One possible explanation for this inconsistency is the differential recruitment of glutamatergic neurons along the rostrocaudal axis of the PPN: studies have shown that activation of caudal PPN neurons promotes exploratory, low-speed locomotion, whereas activation of rostral PPN neurons does not initiate movement and instead suppresses ongoing locomotion[23,32,33,35,37,40,41].

Although the brainstem provides the dominant command to the spinal CPG, other descending pathways, including rubrospinal and

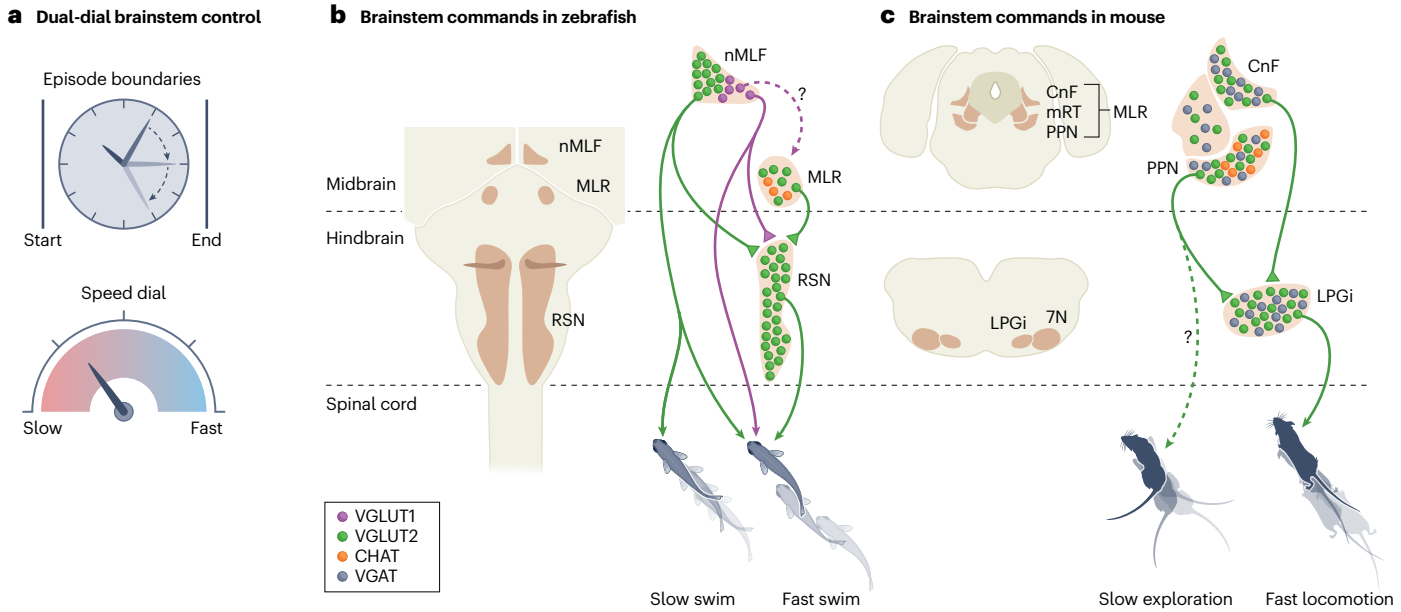


Fig. 2 | Brainstem command circuits as dual dials for initiation and scaling of locomotion. a, Brainstem command circuits operate as a dual-dial control system, providing flexible and combinatorial control of locomotion. One circuit (the 'boundaries dial') defines the duration and maintenance of locomotor episodes, whereas a distinct circuit (the 'speed dial') regulates the intensity and speed of movement. **b**, In zebrafish, brainstem command centres control the initiation, duration and speed of locomotor episodes. The nucleus of the medial longitudinal fasciculus (nMLF) in the midbrain contains two distinct glutamatergic populations expressing vesicular glutamate transporter 1 (Vglut1) and Vglut2. Vglut2$^+$ neurons encode the onset and duration of slow, exploratory swimming, whereas Vglut1$^+$ neurons are recruited at higher speeds to encode fast swimming[43]. Another command centre resides in the mesencephalic locomotor region (MLR)[15,20,23]. Both the nMLF and MLR project to reticulospinal neurons (RSN) in the medulla, with the nMLF also sending direct inputs to the spinal cord (confirmed connections (solid arrows) and unconfirmed connections (question marks and dashed arrows))[34,43]. **c**, In mice, the MLR comprises three nuclei: the pedunculopontine nucleus (PPN), cuneiform nucleus (CnF) and mesencephalic reticular formation (mRT)[20,23]. These nuclei project to the lateral paragigantocellular nucleus (LPGi) in the medulla. Each region contains intermingled inhibitory (vesicular inhibitory amino acid transporter (VGAT)-expressing) and excitatory (VGLUT2$^+$) neurons. The PPN also includes cholinergic (choline *O*-acetyltransferase (CHAT)-expressing) neurons, distinguishing it from the other MLR components[20,23–25]. Functionally, the PPN controls slow, exploratory locomotion, whereas the CnF drives fast, escape-like locomotor movements via activation of LPGi neurons[20]. Parts **b** and **c** are adapted from ref. 15, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

corticospinal tracts, interact extensively with reticulospinal outputs in mammals. These parallel systems underpin the richer voluntary and adaptive repertoire of terrestrial locomotion[12,20,23]. For example, in cats, many spinal interneurons integrate convergent input from reticulospinal, rubrospinal, corticospinal and proprioceptive afferents originating from the periphery, illustrating the integration of descending and sensory control systems that is required for more complex control of limb-driven locomotion[48–50].

These findings support a 'dual-dial' model of brainstem control, in which one set of circuits (the 'boundaries dial') defines the duration of locomotor episodes, whereas another set (the 'speed dial') regulates the speed of movement (Fig. 2a). Each 'dial' thus represents a distinct circuit whose activation scales a specific locomotor parameter. In zebrafish, these dials are mapped onto specific cell types within the nMLF: medial Vglut2$^+$ neurons act as the boundaries dial, whereas lateral Vglut1$^+$ neurons serve as the speed dial. In mice, the caudal PPN probably sets the boundaries of slow exploration, whereas the CnF can act as a speed dial to drive high-speed, escape-like behaviour. Together, these circuits provide flexible and combinatorial control of locomotion.

Defining direction and trajectory

In addition to regulating the 'when' and 'how fast' of movement, locomotion also requires precise steering to align body trajectory with environmental demands. Directional control relies on lateralized brainstem circuits that modulate left–right asymmetries in spinal motor output. In both zebrafish and mice, CHX10-expressing excitatory neurons are key contributors to this function. In mice, unilateral activation of CHX10$^+$ neurons in the gigantocellular reticular nucleus (Gi) induces ipsilateral axial and limb movements, thereby biasing locomotion in that direction[51–55]. These Gi CHX10$^+$ neurons are located downstream of the MLR in the caudal brainstem and project predominantly to the ipsilateral spinal cord. They receive spatially organized inputs from distinct sensorimotor areas, enabling dynamic steering based on context. Within the Gi CHX10$^+$ neuronal population, at least two functionally distinct subsets have been identified: one that projects to the lumbar spinal cord and modulates locomotor speed (and can even induce locomotor arrest) and another that projects to the cervical spinal cord and controls head orientation[53]. Activation of the latter is sufficient to redirect locomotor trajectory by establishing asymmetric changes in head orientation during turning movements. Thus, a population of genetically defined neurons can support several distinct locomotor functions based on their connectivity and activation.

Much of the current evidence for this model in mice derives from gain-of-function studies using optogenetic activation, and whether these CHX10$^+$ subsets are selectively engaged during spontaneous, ethologically relevant behaviours, therefore, remains unclear.

By contrast, the recruitment patterns and functional roles of Chx10$^+$ brainstem neurons have been more precisely resolved in zebrafish. In adult zebrafish, two distinct brainstem circuits outside the MLR and nMLF control directionality. High-amplitude, rapid turns are mediated by a circuit of electrically coupled excitatory Chx10$^+$ and inhibitory commissural Dbx1$^+$ neurons that are distributed across multiple brainstem nuclei[56,57]. These neurons are synchronously activated by pretectal visual inputs and scale their output with the amplitude of the turning angle required. By contrast, slow exploratory turns recruit a separate, spatially restricted subset of Chx10$^+$ neurons that support fine-grained directional adjustments during ongoing swimming, such as those needed during foraging[57]. Together, these findings show that steering is implemented through parameter-specific circuits across vertebrates and illustrate the differences in the degree of mechanistic resolution currently possible in zebrafish and mammals.

Locomotor CPGs

Although locomotor commands are initiated and modulated by the brainstem, their execution is ultimately carried out by spinal circuits. Central to our understanding of this transformation is the concept of the CPG, which has been pivotal in decoding the architecture and logic of vertebrate locomotion.

The emergence of the CPG concept

An early debate in the motor control field centred on whether locomotion rhythm arises from reflex chains (as proposed by Sherrington)[58] or intrinsic spinal mechanisms (as argued by Brown)[59] (Fig. 3a). This question was experimentally addressed through studies showing that rhythmic, alternating activity between flexor and extensor muscles can occur even in the complete absence of sensory input[59]. These findings led to Brown's formulation of the half-centre hypothesis, which proposed that spinal circuits alone are capable of generating basic rhythmic patterns of movement[59,60]. A later revision of this model suggested that, although flexor–extensor alternation is centrally generated, sensory feedback is essential for shaping this activity into the precise and complex patterns observed during natural locomotion[60]. However, subsequent experiments eventually confirmed that the spinal cord is fully capable of producing well-timed and coordinated locomotor patterns independently of peripheral feedback[61,62]. These results laid the foundation for the concept of the CPG as a self-sufficient spinal network that can generate rhythmic, patterned motor output without sensory input. Sensory input was, however, shown to be needed for the continuous adaptation of locomotor movements[12,30,63–65]. Today, this concept has evolved into a dynamic model in which central and sensory mechanisms operate in concert (Fig. 3a). The CPG establishes the

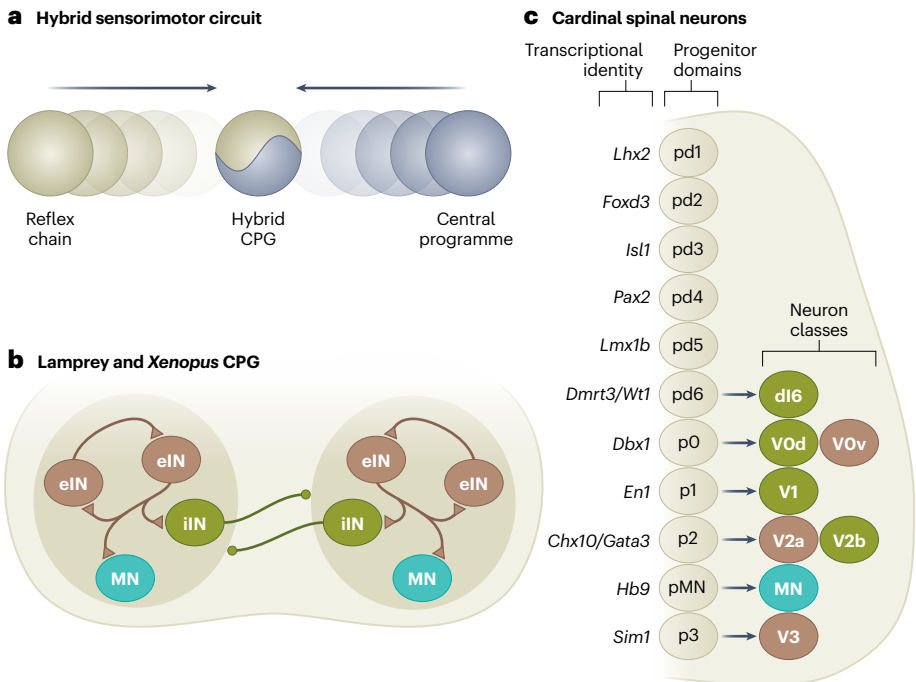


Fig. 3 | The spinal locomotor CPG: from abstract concept to defined circuit architecture. a, Models of locomotor control have shifted from dichotomous frameworks, reflex-chain versus central command, to a unified sensorimotor hybrid central pattern generator (CPG). An early theory proposed that locomotion arises from chained reflexes in which sensory feedback drives successive cycles[58]. An opposing view held that central programmes alone can generate rhythmic activity without feedback[59]. These models have been superseded by a hybrid CPG model in which spinal circuits generate the locomotor rhythm while being continuously shaped by sensory feedback. **b**, Canonical CPG architectures derived from non-mammalian model organisms such as lamprey and *Xenopus* tadpoles illustrate the conserved circuit motifs that underlie rhythm generation and left–right alternation in vertebrates[11,16,69]. Ipsilaterally projecting excitatory interneurons (eINs) generate the rhythm through mutual excitation and drive both motor neurons (MNs) and commissural inhibitory interneurons (iINs). Alternation arises through reciprocal inhibition: when the ipsilateral side is active, its commissural iINs inhibit all CPG neurons on the contralateral side. As the drive from the ipsilateral side subsides, the contralateral side is released from inhibition and becomes active, leading to a phase switch. **c**, The canonical locomotor CPG circuits illustrated in part **b** arise from distinct progenitor domains in the developing spinal cord that give rise to genetically defined cardinal neuronal populations. Neurons involved in locomotor rhythm and pattern generation (including dI6, V0d, V0v, V1, V2a, V2b and V3 interneurons, and MNs) are located in the ventral spinal cord[75–77]. Excitatory populations are shown in brown and inhibitory populations in green.

core rhythm and pattern, whereas sensory feedback tunes transitions between locomotor phases, stabilizes output and adjusts locomotor movements in real time. Furthermore, it has been proposed that, across evolution, the relative contribution of intrinsic spinal CPG mechanisms may decrease as descending and sensory influences become increasingly dominant, particularly in limbed vertebrates[17].

Vertebrate CPG — early lessons

What constitutes the CPG, and how does it generate complex locomotor sequences? Answering these questions requires access to the activity of individual neurons during ongoing locomotion — historically, an impossible requirement to satisfy in mammalian in vivo model systems. Instead, non-mammalian model organisms, such as the lamprey and *Xenopus* tadpoles, provided the necessary accessibility to uncover the basic building blocks of the spinal CPG and their connectivity[11,12,16,19,66–74]. Studies using these organisms have identified the major interneuron types that shape the rhythmic activity of motor neurons and coordinate left–right alternation during swimming (Fig. 3b). Excitatory interneurons with recurrent connections were shown to generate the locomotor rhythm. These interneurons then drive both motor neurons and commissural inhibitory interneurons, which control the alternation of locomotor activity across the midline. These studies have provided the first outline of the structure and function of the vertebrate CPG. The lamprey and *Xenopus* tadpole CPGs remain the most thoroughly understood outside zebrafish. These studies have also highlighted the essential roles of the intrinsic membrane properties of the CPG constituent neurons, their synaptic dynamics and neuromodulatory systems in shaping the locomotor output. Although the CPG underlying swimming with coordinated left–right alternation is now relatively well characterized, particularly in zebrafish (see below), the CPG architecture supporting limbed locomotion remains elusive owing to its increased complexity and the more limited access to identified neurons in vivo.

Molecular era of the CPG

The advent of molecular tools, enabling the selective expression of reporters (such as green fluorescent protein (GFP)), actuators (such as channelrhodopsin-2 (ChR2)) and toxins (such as diphtheria toxin), ushered in a new era in the study of the vertebrate locomotor CPG. These molecular tools revealed that the function of the CPG relies on diverse subpopulations of spinal neurons — each with a specialized function, with motor neurons serving as the final common pathway for muscle activation[7–9,12,13,15,28,70,75]. Crucially, this neuronal diversity is not random but emerges from precise spatiotemporal gene expression programmes that govern the identity, positioning and connectivity of spinal neurons during development.

Early in vertebrate development, morphogen gradients pattern the spinal cord along its dorsoventral axis, giving rise to discrete progenitor domains[8,9,12,13,76–78] (Fig. 3c). Each domain produces a cardinal class of interneurons — such as V0, V1, V2a and V3 interneurons — whose identity is defined by unique transcriptional codes[28,31,75–77,79,80]. This spatial patterning is complemented by temporal patterning: neurons born at different developmental stages express stage-specific transcription factors that further refine their identity[28,29,78,80]. In the ventral spinal cord, early-born 'Z-group' neurons typically occupy lateral positions and extend long-range projections, whereas later-born 'N-group' neurons are more medially positioned and form local circuits[28,78]. These orthogonal spatial and temporal programmes create a layered molecular logic that guides circuit assembly.

The broad spatiotemporal classification of neurons seen during development is further refined in the mature spinal cord. Interestingly, although each population is molecularly distinct, there is also a degree of molecular overlap between neurons from different populations[29,75,79,80]. This suggests that neurons within the locomotor CPG may share some molecular characteristics while maintaining functional specialization. In mammals, the mapping of molecular identity, connectivity and functional subtype remains far less resolved, and most developmentally defined interneuron subclasses have not yet been subdivided according to their speed-dependent recruitment or connectivity. Although studies in mice have begun to reveal that specific neuronal populations are linked to discrete aspects of locomotor control[5,7,8,81], direct evidence for a molecular–functional correspondence comparable to that observed in adult zebrafish is still limited.

These findings show that the locomotor CPG is built from developmentally specified neuronal populations. Recent findings suggest that these neuronal populations are organized into speed-specific 'modules', a term that I use here to refer to discrete functional circuits composed of specific subtypes of neurons (interneurons and motor neurons). Each module is defined by three criteria: preferential recruitment of the neurons within the module at a specific locomotor speed, enriched connectivity among neurons within the module (relative to those engaged at other speeds), and shared intrinsic and synaptic properties that support coordinated activation within that speed range. The next sections examine how this organizational logic gives rise to functionally distinct spinal subcircuits dedicated to locomotor rhythm ignition, speed control and pattern coordination.

Entry point to the locomotor CPG

Locomotor initiation depends on excitatory interneurons that serve as the entry point to the spinal CPG, transforming descending commands from the brainstem into patterned activity that defines movement strength, speed and timing. Through their intrinsic membrane properties and defined synaptic connectivity, these neurons activate the locomotor CPG and shape motor neuron output. Identifying such 'entry point' neurons is essential for understanding how locomotor commands are generated and flexibly modulated. Although this has been achieved in zebrafish, their identity in mammals remains unresolved.

Single entry point in zebrafish

In zebrafish, the excitatory core of the spinal CPG is composed of V2a interneurons, defined by expression of Chx10 (also known as Vsx2)[15,39,46,82–85]. Targeted laser ablation and optogenetic manipulation have shown that V2a interneurons are both necessary and sufficient for the initiation and maintenance of rhythmic swimming[85]. Acute ablation of V2a neurons precludes the transformation of brainstem commands into spinal rhythmic output, abolishing swimming. Conversely, optogenetic activation of these neurons elicits rhythmic, coordinated locomotor activity in both ex vivo and in vivo settings[15,84]. These findings have established V2a interneurons as the functional entry point to the spinal CPG, converting descending signals into motor patterns that propagate through the spinal network (Fig. 4a).

Multiple entry points in mice

In mice, the identity of the excitatory interneurons in the CPG that initiate locomotion remains unresolved. Several populations have been implicated, including V2a interneurons, interneurons defined by expression of transcription factors such as HB9, SHOX2 and LHX9, and ventral spinocerebellar tract (VSCT) neurons[7,86–93]. However, none of

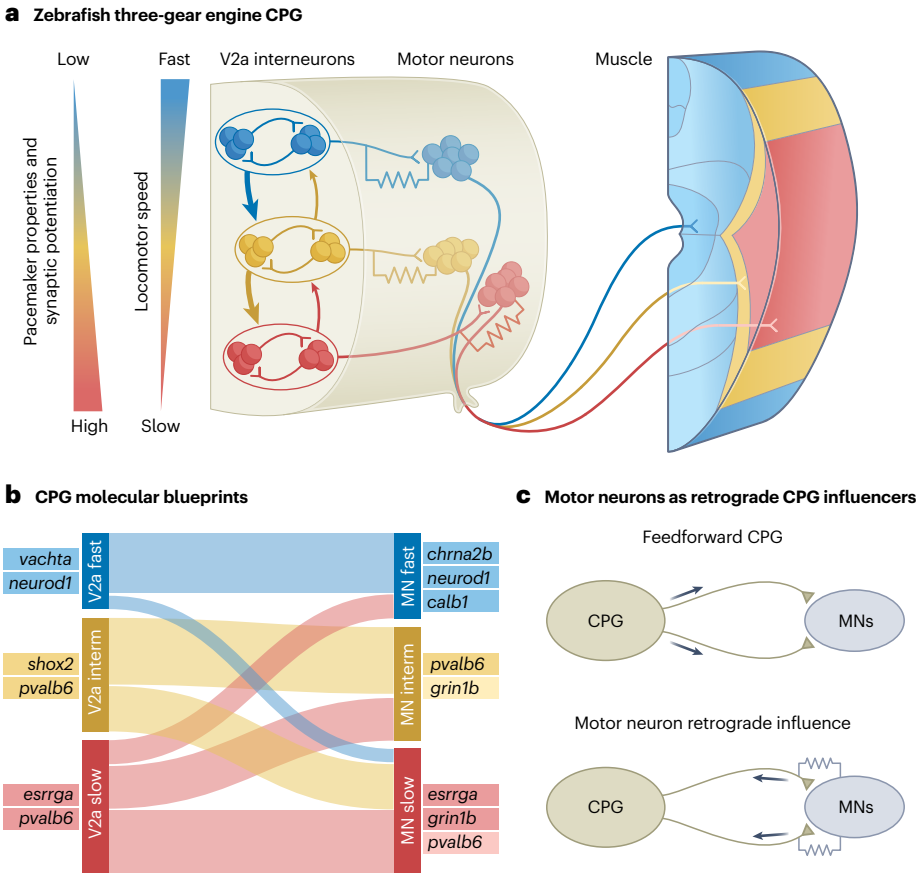


Fig. 4 | A modular, three-gear locomotor CPG built from molecularly defined circuit units. a, In adult zebrafish, the spinal locomotor central pattern generator (CPG) is organized into three discrete, speed-tuned modules – slow (red), intermediate (yellow) and fast (blue) – each comprising a specific set of V2a interneurons and their target motor neurons (which innervate slow, intermediate and fast muscle, respectively)[15,46]. Modules are recruited sequentially as locomotor speed increases. The slow module contains neurons with stronger pacemaker properties and greater synaptic potentiation, features that enable its early recruitment[85,97]. Faster modules subsequently exert top–down control over slower ones via asymmetrical synaptic connections that are stronger and more frequent from fast to slow modules[85]. **b**, Each circuit module is underpinned by neurons with distinct transcriptional identities. V2a interneurons and their target motor neurons within a given module share module-specific gene expression profiles, with the fast module showing the most divergent molecular signature[118]. Line thickness connecting V2a interneurons and motor neurons indicates the degree of transcriptional overlap, which is higher within modules than between modules. The listed genes represent key markers of V2a interneuron and motor neuron subtypes, with box shading reflecting relative enrichment. Together, these features reveal a molecular blueprint that mirrors functional organization and recruitment logic. **c**, Traditionally, motor neurons have been viewed as passive output elements of the locomotor CPG, relaying pre-structured motor patterns to muscles in a strictly feedforward manner. However, evidence shows that motor neurons can exert retrograde influence on the CPG via gap junctions, enabling them to modulate ongoing rhythmic activity[162]. Through these electrical synapses, motor neurons gain access to core rhythm-generating circuitry and actively shape locomotor output. In this revised framework, motor neurons are not merely end points of motor execution but are integrated into the CPG as bona fide circuit members. Part **a** is adapted from ref. 15, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). Part **b** is adapted from ref. 118, Springer Nature Limited.

these populations has been shown to be both necessary and sufficient for locomotor initiation, as is the case for V2a interneurons in zebrafish.

In newborn mice, genetic ablation of V2a interneurons impairs sensory-evoked and brainstem-evoked locomotion but fails to abolish drug-induced locomotor rhythm, although left–right alternation is lost[89,90,94]. Consistent with this, ablation of V2a interneurons in adult mice leads to a speed-dependent change in locomotor gait from trot to gallop, suggesting that their primary role is in left–right coordination rather than rhythmogenesis[89,90,94].

Similarly, genetic ablation or silencing of HB9-expressing, SHOX2-expressing or LHX9-expressing interneurons yields only modest disruption of drug-induced locomotor-like activity in neonatal spinal cord preparations[7,8,12,13]. By contrast, VSCT neurons, whose molecular identity is still unclear, have emerged as a more promising candidate for the CPG entry point. Their activation induces locomotor-like rhythms, and their silencing impairs sensory-evoked locomotion[86]. However, VSCT neurons are electrically coupled to motor neurons, and their silencing, therefore, probably reduces motor neuron excitability. This makes it difficult to determine whether VSCT neuron silencing disrupts the rhythm generator itself or merely suppresses its downstream output. Thus, current evidence does not yet allow us to conclude that VSCT neurons constitute a dedicated rhythm-generating population. Overall, the mechanistic contribution of each candidate population as a potential rhythm-generating entry point to the CPG in mice remains

unresolved, reflecting both the complexity of the mammalian spinal cord and the limited ability to monitor and manipulate identified neurons during natural movement.

The entry point to the locomotor CPG in mice, therefore, appears more complex than that of zebrafish, probably relying on multiple, partially redundant excitatory interneuron populations. This may be because limbed vertebrates require distributed entry points to coordinate forelimb–hindlimb, flexor–extensor and left–right dynamics, whereas axial swimmers can rely on a single, longitudinally organized excitatory drive. That the ablation of one population of interneurons in mice alters specific aspects of rhythm generation without entirely abolishing locomotion supports this idea.

CPG organizational logic for ignition and speed control

How do the excitatory interneurons that serve as entry points to the CPG generate rhythmic locomotor activity, and how do they regulate speed? Recent studies in adult zebrafish have provided insights into the relative contributions of recurrent excitatory loops between V2a interneurons, their hierarchical interactions with each other, and their intrinsic pacemaker properties to locomotor rhythm generation at different speeds. Together, these mechanisms determine when a rhythm begins (ignition), how stable it is, and how it scales with speed[95]. Together, these findings have redefined CPG organization, shifting from the classical view of a unitary network to a multilayered, three-gear architecture composed of discrete speed-specific CPG circuit modules.

Excitatory circuits — the engine

Dual patch clamp recordings between V2a interneurons in adult zebrafish have been used to define the synaptic connectivity rules and intrinsic properties that support rhythm initiation and speed encoding in the CPG[39,85,96,97] (Fig. 4a). These studies have revealed that V2a interneurons are functionally heterogeneous, comprising three physiologically distinct subtypes (slow, intermediate and fast), each tuned to a specific locomotor speed[39,82,85,97] (Fig. 4a). These three subtypes form discrete, speed-specific excitatory circuits, each comprising V2a interneurons that are mutually connected and that form recurrent excitatory circuits that can act as autonomous rhythm-generating units (Fig. 4a). These three speed-tuned circuits, thus, operate as a 'three-gear engine', with each gear initiating and maintaining rhythmic activity within its preferred speed range. To date, the existence of this three-gear architecture has been established primarily in adult zebrafish: whether an equivalent organization exists in mammals remains unknown.

The work in zebrafish has further shown that transitions between speeds, akin to 'gear shifts', are mediated by asymmetric connectivity between the speed-specific circuits (Fig. 4a). Faster circuits make more frequent and stronger connections onto slower ones than vice versa[85]. This arrangement allows the activity of the faster circuit to override and entrain the slower circuit as locomotor speed increases. Locomotion is typically initiated by the slow circuit, but as excitation builds, the intermediate circuit imposes a faster rhythm on the circuit. At the highest speeds, the fast circuit dominates, propagating its rhythm downward through the hierarchy. During deceleration, the circuits are gradually de-recruited in reverse order. This layered hierarchical organization ensures smooth transitions and graded speed control. Each circuit operates within its optimal frequency range, whereas asymmetric top–down connectivity prevents interference between circuits. This mechanism also provides circuit-level robustness, allowing the system to respond flexibly to varying locomotor demands.

In addition to recurrent connectivity, rhythmogenesis in the zebrafish CPG relies on conditional pacemaker properties in a subset of V2a interneurons that endow them with the ability to transform tonic excitation into rhythmic bursting[15,46,85,97]. These neurons act as local ignition points that initiate CPG activity. Pacemaker V2a interneurons are most abundant in the slow circuits, wherein they support the initiation of locomotion when descending brainstem drive is weak[85,97]. As speed increases, pacemaker interneurons in the intermediate and fast circuits are sequentially recruited. The fast circuit contains far fewer pacemaker V2a interneurons than the slow one, suggesting that rhythm at high speeds is driven primarily by network-level recurrent excitation among interconnected V2a interneurons rather than intrinsic pacemaker mechanisms[85]. This suggests that rhythm generation shifts from being pacemaker-driven at low speeds to network-driven at high speeds, reflecting a synergistic strategy that balances reliable initiation with scalable output. Pacemaker-like properties in some excitatory interneurons and motor neurons have also been described in mice, particularly in neonatal or in vitro preparations, and may contribute to the ignition of spinal locomotor CPG activity[95,98–100].

Together, these findings suggest that these speed-tuned excitatory circuits form a three-gear engine (Fig. 4a). This design, combining recurrent excitation with embedded pacemaker neurons, ensures robust ignition and flexible speed control across a wide dynamic range of locomotor outputs. However, this framework currently rests on the high-resolution circuit dissection achievable in adult zebrafish; whether mammals share this organizational logic or instead rely on distinct mechanisms suited to limbed biomechanics remains a major open question.

A modular CPG–motor neuron architecture

The organization of rhythm-generating V2a interneurons offers a powerful framework for understanding how spinal circuits regulate locomotor speed. However, how is this structured activity transformed into coordinated motor output? Recent work reveals that the organizational logic of the CPG extends to motor neuron recruitment, linking distinct V2a subtypes to the orderly, speed-tuned activation of motor neuron pools[82,97,101,102].

Three circuit modules — the gearshift. In zebrafish, dual whole-cell recordings between V2a interneurons and motor neurons have uncovered a circuit architecture composed of three discrete, speed-tuned modules (Fig. 4a). Each module consists of a V2a subtype that forms targeted synaptic connections with a corresponding motor neuron subtype (pool) — slow, intermediate or fast — and there is minimal cross-connectivity between modules[82]. Furthermore, the V2a interneuron and motor neuron subtypes that are connected together form matched pairs that are recruited together during locomotion and possess complementary intrinsic and synaptic properties[85,97]. V2a interneurons and motor neurons in the slow module have low firing thresholds and exhibit strong, facilitating NMDA receptor-dependent synaptic transmission, enabling early and sustained activation at low speeds. By contrast, fast module V2a interneurons and motor neurons display high firing thresholds and linear, non-facilitating synapses, requiring greater drive for recruitment and confining their activation to high-speed locomotion. Intermediate module properties fall between these two extremes. This architecture challenges classical models that envision the locomotor CPG as a unitary excitatory network that modulates speed by uniformly scaling excitatory drive, akin to pressing harder on a throttle. Instead, it suggests that speed is regulated through

the selective engagement of discrete V2a interneuron–motor neuron circuit modules, each optimized for a specific locomotor regime.

The modular organization uncovered in zebrafish endows the spinal locomotor circuits with an intrinsic gearshift mechanism, whereby speed increases are achieved through the incremental engagement of distinct motor units. Mammals also recruit motor units in a slow-to-fast sequence as locomotor speed increases, but the underlying circuit connectivity remains undefined. Mammalian motor pools typically contain a mixture of muscle fibre types and corresponding motor neuron subtypes[81,103–105]. It is possible that the slow, intermediate and fast motor neurons within each mammalian motor pool could, like those of zebrafish, form discrete circuit modules, each driven by a distinct class of excitatory premotor interneurons. Such a modular circuit organization would not only support the orderly recruitment of motor units but also enhance the flexibility to selectively coordinate slow, intermediate and fast motor units across synergistic pools, thereby optimizing motor synergies during locomotion. However, this remains a working hypothesis, and direct experimental evidence for such speed-specific interneuron–motor neuron modules in mammals is still lacking.

Beyond the size principle. The modular organization revealed in zebrafish challenges the size principle, the canonical rule that posits that all motor neurons receive a common excitatory drive and are recruited in a fixed order based on their input resistance and threshold[106,107]. In larval zebrafish, this recruitment logic appears to hold[108]. However, during development, the spinal CPG undergoes extensive anatomical and functional remodelling, reshaping motor neuron recruitment dynamics (Box 1). In adult zebrafish, motor neurons with similar input resistances receive distinct, subtype-specific excitatory input from V2a interneurons, undermining the assumption of uniform drive[82]. Moreover, their recruitment is dictated not by passive properties but by their module-specific connectivity, membrane properties and synaptic dynamics[85,101,102]. Thus, the size principle gives way to more flexible, circuit-defined recruitment mechanisms that allow precise and context-dependent control of motor output.

Molecular blueprint for circuit modules. What governs the assembly of these circuit modules? Do matching molecular identities guide the pairing of interneurons and motor neurons? Traditional marker-based analyses, such as those revealing the diversity of V1 interneurons in mice, have long suggested that molecular heterogeneity may underlie circuit specificity[109,110]. Single-cell RNA sequencing has elaborated upon this view, identifying distinct transcriptional signatures among motor neuron subtypes, including alpha, beta and gamma motor neurons, as well as among motor neurons targeting specific muscles or muscle fibre types[28,29,79,111–117].

In adult zebrafish, transcriptomic analyses have provided compelling evidence that molecular identity parallels functional organization (Fig. 4b). Motor neurons cluster into three transcriptionally distinct subtypes that innervate slow, intermediate and fast muscle fibres[15,118]. Fast motor neurons exhibit the most divergent gene expression profiles, whereas slow and intermediate types show partial overlap. Strikingly, zebrafish larvae display only two transcriptional motor neuron clusters, corresponding roughly to fast and slow or intermediate types[119], suggesting that intermediate identity emerges later, potentially shaped by circuit refinement and feedback from muscle activity.

A matching tripartite molecular logic is observed in adult V2a interneurons. Single-cell RNA sequencing reveals three transcriptionally defined V2a subtypes, each characterized by a specific set of enriched genes[118]. Combining genetic and electrophysiology analyses, it was shown that these transcriptionally defined V2a interneuron subtypes align with their recruitment at slow, intermediate or fast locomotor speeds[15,118]. These subtypes are specified early in development[119] and exhibit distinct transcription factor profiles that probably govern their excitability, synaptic properties and connectivity.

Crucially, V2a interneurons and their postsynaptic motor neuron partners within each module share overlapping transcriptional signatures[118]. This alignment suggests a molecular logic in which the same transcriptional programmes that define neuronal identity also specify synaptic pairing and circuit assembly (Fig. 4b). Rather than being merely compatible, V2a interneurons and their motor neuron targets within each module are transcriptionally matched to function as cohesive units. In mammals, evidence for functionally matched interneuron–motor neuron transcriptional subtypes remains sparse, and whether a comparable tripartite molecular logic exists is unknown.

Together, these findings support a model in which spinal circuits are organized along two intersecting molecular axes: one that specifies broad neuronal class (interneuron versus motor neuron, for example) and another that defines functional module (slow, intermediate or fast). This orthogonal framework offers a scalable, genetically encoded blueprint for modular circuit design — one that is firmly supported in adult zebrafish and provides testable hypotheses for conserved principles across other vertebrate species.

CPG coordination of alternation and timing

Locomotor movements unfold stepwise, with each step or body bend characterized by alternating neuronal activity across the two sides of the body. A hallmark of locomotion is this precise left–right coordination, which in limbed animals extends across multiple joints and adapts with speed and gait. In addition to left–right alternation, precise flexor–extensor coordination is required to control the timing of muscle activation around individual joints. To achieve this coordination across speeds and locomotor modes, the excitatory drive from CPG is sculpted into coordinated left–right and flexor–extensor patterns by both inhibitory and excitatory spinal interneurons, which act as timing regulators and patterning elements within the CPG.

Commissural interneurons mediate CPG alternation

Commissural interneurons within the spinal cord are key for alternation[8,12,13,46,120,121]. They comprise both inhibitory interneurons that promote alternation through mid-cycle inhibition and excitatory interneurons that promote synchrony. Early pharmacological studies in both non-mammalian and mammalian models have shown that blocking inhibition converts the normal left–right alternating motor pattern into a synchronous, slower rhythm that is driven by the activity of excitatory commissural interneurons[122–125].

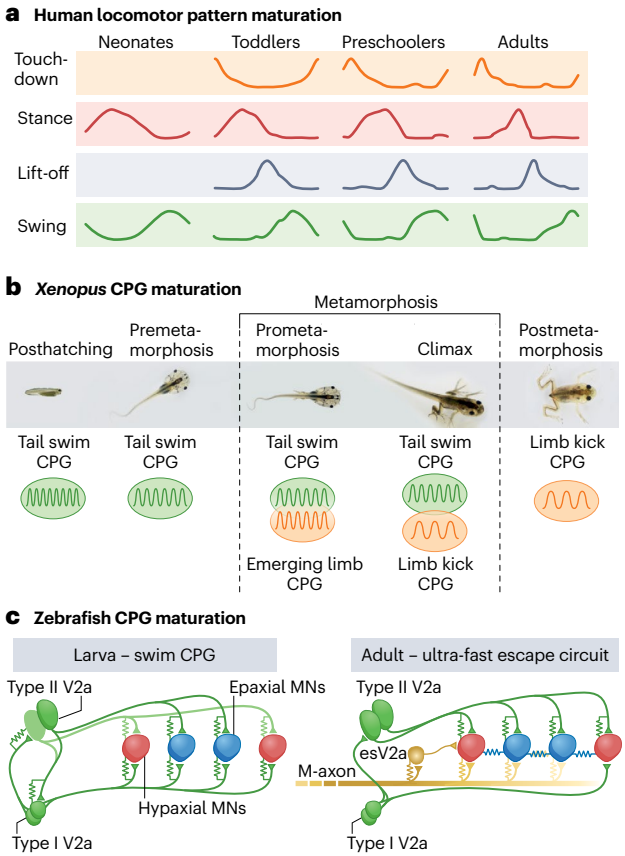
Among commissural interneurons, the V0 population, defined by expression of the transcription factor DBX1 (ref. 126), is critical for left–right coordination (Fig. 5a). Deletion of this transcription factor or ablation of this neuronal population abolishes left–right alternation, resulting in synchronous activity on both sides of the spinal cord[127–130]. This population comprises inhibitory V0d and excitatory V0v interneurons, each of which makes distinct contributions to locomotor control (Fig. 5a). In mice, genetic perturbations and model-based interrogations suggest that V0d and V0v neurons operate in a dual-mode fashion to control gait changes in a speed-dependent manner: inhibitory V0d neurons support alternation at lower speeds and the walking gait,

Box 1 | Transformation of the CPG during development

The locomotor central pattern generator (CPG) does not emerge fully formed at birth but develops through a dynamic interplay between genetic programming, neural plasticity and environmental influences[19,185–188]. In some species, spinal circuits capable of generating patterned motor activity are functional at birth; in other species, substantial postnatal maturation is required before independent locomotion can occur. For example, human newborns display a primitive stepping reflex consisting of two basic phases (stance and swing). This temporarily disappears, only to re-emerge with assisted walking and later mature into four-phase mature cycle (stance, touch-down, swing, lift-off), reflecting increased temporal precision and sensorimotor integration (see the figure, part **a**).

One striking example of CPG transformation occurs in *Xenopus* frogs. Larval swimming is driven by an axial CPG (see the figure, part **b**). During metamorphosis, a limb-based CPG gradually emerges[19] and, for a transitional period, operates in parallel with the immature axial CPG. At metamorphosis climax, the axial system is silenced or repurposed, leaving mature limb-based locomotion dominant[19,189]. Thus, new motor networks can be layered onto pre-existing ones and dynamically modulated into coherence or independence according to behavioural demands. By contrast, salamanders retain both axial and limb locomotor versatility into adulthood, enabling switches between tail-based swimming and limb-based walking[17,190,191]. Whether these gaits are governed by distinct CPGs or a reconfigurable, shared circuit remains unresolved, but descending modulatory input and sensory feedback are central to gait selection and refinement[192].

The fish locomotor CPG also undergoes pronounced remodelling. Zebrafish larvae perform high-frequency burst-and-glide swimming driven by a fast CPG composed of early-born primary motor neurons and fast V2a interneurons[193,194] (see the figure, part **c**). Adults display slower, more continuous swimming, enabled by the sequential recruitment of fast, intermediate and slow spinal CPG circuits. In parallel, the larval fast circuit is repurposed through synaptic pruning, and new connectivity becomes specialized for escape[161,195]. This ultra-fast escape system involves specialized cholinergic escape V2a interneurons (esV2a) and ultra-fast primary motor neurons innervating epaxial and hypaxial fast muscles. Thus, developmental remodelling can convert a multifunctional immature CPG into a task-specific mature circuit. Sensory feedback systems also remodel; for example, Rohon–Beard neurons degenerate in late larvae and are replaced by an intraspinal proprioceptive organ in juveniles and adults[15,175].



These examples show that the locomotor CPG is not a static network but a developmentally dynamic scaffold. Through layering, pruning and specialization, circuits are progressively reconfigured into diverse locomotor repertoires, shaped by intrinsic developmental programmes and external demands. Part **a** is adapted with permission from ref. 186, Elsevier. Part **b** is adapted from ref. 19, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). Part **c** is adapted with permission from ref. 161, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

whereas excitatory V0v neurons maintain alternation during moderate to high speeds and promote the trot gait[128,129] (Fig. 5b,c). However, direct physiological evidence for their speed-dependent recruitment and connectivity remains limited. Recent work in adult zebrafish has started to fill this gap, providing the first direct physiological recordings from V0d subpopulations across speeds and showing that V0d interneurons are preferentially active at slow speeds, whereas V0v neurons dominate during faster swimming[131,132]. Although zebrafish do not exhibit the discrete gait transitions observed in mice, these findings support a conserved division of labour between V0d and V0v interneurons[128,129,131,132].

In accordance with this idea, ablation of V0d neurons in adult zebrafish impairs slow, exploratory swimming and leads to a loss of mid-cycle inhibition in motor neurons. Interestingly, the V0d interneuron recruitment pattern is reversed in larval zebrafish, in which V0d interneurons are selectively recruited during escape and high-frequency swimming[132–135], whereas a distinct V0v subtype, known as MCoDs, is recruited during slow swimming[96,108,133–137]. These findings point to developmental shifts in the roles of commissural interneurons and raise questions about how speed-dependent alternation strategies evolve across life stages and across species.

A second class of inhibitory commissural interneurons, the dI6 population (marked by expression of the transcription factor DMRT3), also contributes to locomotor coordination across species. In larval zebrafish, dI6 neurons are essential for alternation at slow speeds[134,138]. In mammals, dI6 neurons have a role in gait coordination: mutation in

the *DMRT3* gene impacts gait control in Icelandic horses[139], whereas these interneurons contribute to regulating left–right alternation during drug-induced locomotor activity in mice[139,140].

Finally, V3 interneurons, an excitatory commissural population, have been shown in mice to promote left–right synchrony and stabilize high-speed gaits, whereas zebrafish homologues contribute to the robustness of the locomotor rhythm[141–144]. Although they are not central to alternation, their role in stabilizing output complements the inhibitory–excitatory balance that governs pattern formation and gait control.

Ipsilateral inhibition for CPG timing

In addition to interlimb alternation, locomotion requires the activity of the muscles within each limb or body segment to be precisely timed. This intralimb patterning relies on ipsilateral inhibition – inhibition that originates and acts on the same side of the spinal cord. Among the best-characterized ipsilateral inhibitory populations are the V1 interneurons, a molecularly diverse group[109,110] that gives rise to several functionally defined interneuron subtypes, including Renshaw cells, Ia inhibitory interneurons and Ib interneurons[8,120]. Classical work in the cat has shown that these spinal interneurons are active during locomotion and receive convergent input from reticulospinal, rubrospinal, corticospinal and proprioceptive pathways[145,146]. In neonatal mice, ablation or silencing of V1 neurons slows drug-induced rhythmic activity, suggesting that these neurons contribute to the timing and pacing of the locomotor rhythm[147].

Direct recordings in non-mammalian species, such as *Xenopus* tadpoles and larval zebrafish, have provided deeper mechanistic insight into the roles of V1 interneurons[148–150]. In these animals, V1 interneurons fire in-phase with ipsilateral motor neurons but with a delay, discharging late in the motor burst[148–151]. This delayed inhibition appears to contribute to the termination of the motor burst, effectively setting its duration and regulating locomotion speed. Ablation of V1 interneurons prolongs burst duration and impairs transitions to higher speeds[149], indicating a conserved role for these neurons in timing and burst shaping across vertebrates.

Beyond timing, ipsilateral inhibition is also essential for interjoint coordination. Studies in mice have shown that V1 and V2b interneurons together regulate flexor–extensor alternation[152]. Silencing both populations disrupts alternation between these opposing muscle groups, suggesting that ipsilateral inhibitory circuits are essential for organizing joint-level motor output. Although zebrafish lack limbs and, thus, do not require interjoint coordination, V2b interneurons serve a different role in shaping motor output. Rather than coordinating flexor–extensor alternation, V2b interneurons in larval zebrafish directly inhibit motor neurons and act as a brake during swimming[153,154]. Their optogenetic inhibition elicits faster locomotion, whereas their activation slows locomotion, suggesting an apparent divergence in function from speed control in fish to joint coordination in mammals.

Together, these findings suggest that commissural and ipsilateral inhibitory interneurons serve complementary roles in the CPG: commissural circuits enforce left–right alternation, whereas ipsilateral

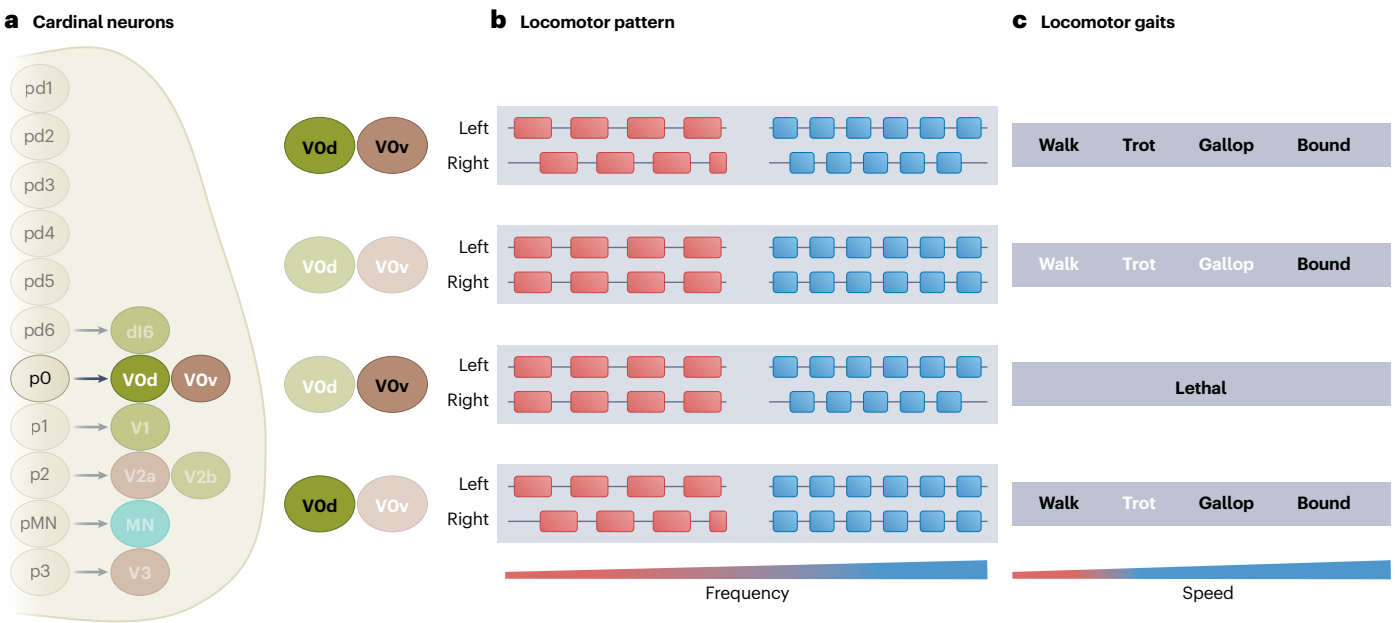


Fig. 5 | Speed-specific coordination of alternation and gait. a, V0 commissural interneurons develop from the p0 progenitor domain in the ventral spinal cord, differentiating into inhibitory dorsal (V0d) and excitatory ventral (V0v) interneurons[126,127]. **b**, In the isolated spinal cord of newborn wild-type mice, locomotor networks maintain robust left–right alternation at both slow and fast frequencies. This coordination is critically dependent on V0 neurons, as ablating the entire V0 population abolishes alternation, leading to left–right synchrony at all frequencies[128]. This control is speed-dependent and subdivided between V0 interneuron subtypes: specific ablation of V0d interneurons disrupts alternation only at slow frequencies, whereas ablation of V0v interneurons preserves slow alternation but results in a synchronous locomotor pattern at fast frequencies[128]. **c**, Adult wild-type mice exhibit a speed-dependent gait transition from walk to trot to gallop and bound. This repertoire is severely restricted in V0 mutants. Mice lacking all V0 neurons lose all alternating gaits (walk, trot, gallop) and exclusively express a bounding gait at all speeds[129]. Although the specific contribution of V0d interneurons to adult gait remains unknown owing to lethality, V0v ablation selectively impairs the trot. These mice retain the ability to walk, gallop and bound, but they cannot execute the intermediate trotting gait[129]. Adapted from ref. 13, Springer Nature Limited.

inhibition regulates timing and joint coordination. These inhibitory systems interact with the aforementioned modular excitatory framework to shape patterned motor output across speeds, gaits and developmental stages. Thus, locomotion relies not only on the excitatory engine that generates rhythmic drive but also on a set of inhibitory circuit maestros that sculpt, pace and coordinate this drive into coherent motor behaviour.

Motor neurons contribute to the CPG

The execution and modulation of locomotor movements ultimately depend on activation of motor neurons. Originally designated by Sherrington as the "final common path", motor neurons have long been viewed as passive relays, transmitting centrally generated motor commands to muscles without shaping those commands (Fig. 4c). According to this traditional view, the only recognized central feedback to the CPG is via Renshaw cells, which are thought to modulate motor output through recurrent inhibition without influencing the rhythm-generating core.

This view is now obsolete. A growing body of evidence across species has shown that motor neurons exert a retrograde influence on premotor (CPG) circuits, positioning them as active contributors to rhythm generation. In the developing chick spinal cord, spontaneous rhythmic bursting in the spinal cord can be initiated by motor neuron activity[155]. Similar findings in mouse embryos demonstrate that motor neuron-derived signals shape early network activity[156]. In neonatal rodents, brief bursts of stimulation of the ventral root (containing motor neurons) entrain rhythmic activity in the spinal cord under disinhibited conditions[157,158]. Furthermore, optogenetic silencing of motor neurons in neonatal mice reduces the frequency of drug-induced fictive locomotion[159]. These effects are thought to be mediated by a retrograde glutamatergic influence from motor neurons onto interneurons[159,160].

A more mechanistic interrogation of the role of motor neuron feedback has been provided by studies in adult zebrafish, in which motor neurons have been shown to retrogradely regulate excitatory premotor V2a interneurons via bidirectional gap junctions[161,162] (Fig. 4c). The connections between interneurons and motor neurons form a loop in which motor neurons are not simply end points but actively shape upstream circuit activity. The electrical component of the mixed chemical–electrical synapses between V2a interneurons and motor neurons allows changes in motor neuron membrane potential to propagate backward into presynaptic V2a axons. There, they modulate local synaptic strength and firing at the soma. Thus, as motor neurons receive rhythmic excitation alternating with inhibition from the CPG during locomotion, they dynamically gate their own upstream drive — potentiating excitatory input during the active (excitation) phase and suppressing it during the inactive (inhibition) phase. In effect, motor neurons have a role in controlling their own excitatory drive, introducing an additional layer of regulation within the CPG architecture. Supporting this regulatory role, optogenetic inhibition of motor neurons during ongoing swimming reduces both the frequency and duration of locomotion.

In the mammalian spinal cord, ultrastructural studies combined with analyses of the expression of connexins (of the proteins that form gap junctions) suggest that a substantial fraction of excitatory connections are mixed synapses[163,164]. These electrical–chemical synapses emerge after the second to third postnatal week and, thereafter, increase to adult levels[163], raising the possibility that mammalian motor neurons, too, can retrogradely influence the CPG — a plausible mechanism that remains to be experimentally tested.

Together, these findings demand a conceptual shift: motor neurons are not merely passive transmitters of motor commands. They exert both forward (to muscle) and backward (to the CPG) influence, making them key elements of the locomotor CPG — bona fide members of the circuits that they were once thought to merely serve (Fig. 4c).

Proprioceptive control of the CPG

Sensory feedback is crucial for both the execution and adaptability of locomotion. At its core is proprioception, the continuous internal monitoring of body position and movement. Proprioception relies on mechanosensation, the conversion of physical forces into neural signals. In vertebrates, this conversion is primarily mediated by PIEZO channels[165], a family of mechanosensitive ion channels that allow the nervous system to detect the physical interactions of the body with the environment[166–169]. Proprioception provides the CNS with online feedback about joint angles, limb trajectories and muscle forces — information that is essential for coordinating movement with precision. When proprioception is disrupted in humans, as a result of either *PIEZO2* mutations or damage to sensory neurons, locomotion becomes awkward, delayed and overly reliant on visual feedback[30,170]. Comparable deficits are seen in mice lacking PIEZO channels in proprioceptive neurons or animals in which proprioceptive afferents are ablated[171].

Peripheral sensory feedback

In addition to providing a baseline map of body position, sensory feedback continuously shapes the rhythm and timing of locomotor output[12,63–65,172] (Fig. 6a). Inputs to the CPG from cutaneous and proprioceptive receptors modulate transitions between locomotor phases and adjust motor output to accommodate external perturbations or obstacles. During walking in vertebrates, two proprioceptive signals are especially important for regulating the transition from stance to swing. First, extension of the hip to a threshold angle (signalled by muscle spindles, the proprioceptors present in hip flexors) triggers swing initiation[12,63–65,173]. Second, the loading of extensor muscles (force exerted owing to the weight bearing), sensed by Golgi tendon organs, delays swing onset until unloading occurs, effectively acting as a brake[12,63–65,173]. Together, these signals ensure that limb transitions are temporally aligned with biomechanical demands during locomotion.

Cutaneous inputs exert phase-dependent effects on locomotion. In spinalized cats, stimulation of the dorsal paw during stance enhances extensor activity, reinforcing weight support. The same stimulus during swing elicits the stumbling corrective reaction — a reflexive elevation of the foot via distal flexor activation to avoid a potential obstacle[12,63–65,173]. This phase-dependent reflex reversal illustrates a key computational feature of spinal circuits. Sensory inputs are not hardwired but are contextually filtered and transformed on the basis of ongoing motor activity. This modulation is gated by the spinal CPG, which alters the transmission of sensory inputs to the CPG[65]. Within the CPG, presynaptic inhibition of sensory synaptic transmission acts as a dynamic filter that allows irrelevant inputs to be suppressed while preserving those aligned with ongoing motor performance[65]. Thus, the CPG is not merely a rhythm generator but also a dynamic filter for incoming sensory signals, sculpting reflex pathways in real time to maintain locomotor stability.

Central proprioception within the CPG

Traditionally, sensory feedback has been viewed as an extrinsic modulator, adjusting a centrally generated motor pattern after the movement has taken place (Fig. 6a). This assumption reflected the view that

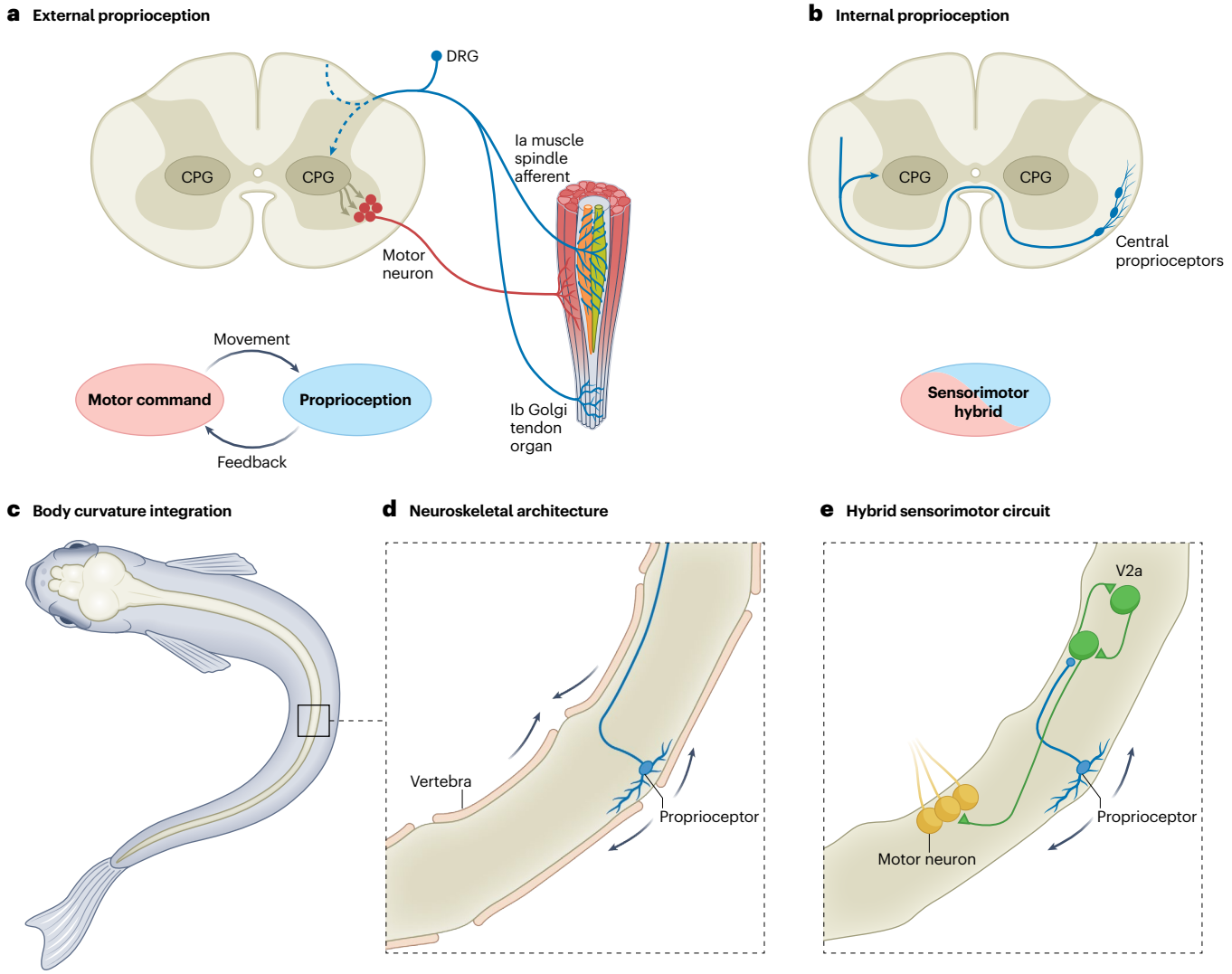


Fig. 6 | Proprioceptive feedback to the CPG. a, Classical models of proprioception describe 'outer proprioceptors', peripheral sensors such as Ia muscle spindle afferents and Ib Golgi tendon organs, that detect the biomechanical consequences of centrally generated motor commands[63]. This feedback arrives at the spinal cord with a delay and modulates the output of an otherwise autonomous CPG, reinforcing a view of sensory and motor systems as sequential and functionally distinct. **b**, Studies in non-mammalian vertebrates (lamprey and zebrafish) have revealed the existence of 'inner proprioceptors', neurons located within the spinal cord that integrate body kinematics with motor command generation in real time[174,175]. These dual-function neurons serve both sensory and motor roles, enabling online modulation of the locomotor programme. This reorganization collapses the sensory–motor dichotomy, giving rise to a hybrid sensorimotor CPG in which proprioception is not a downstream feedback mechanism but an embedded, constitutive component of motor circuits. **c**, During fish locomotion, the body exhibits rhythmic changes in curvature that must be precisely tracked to maintain coordinated movement. **d**, In zebrafish, a population of inner proprioceptive neurons located within the spinal cord express the mechanosensitive channel Piezo2 (blue in part **e**) and are segmentally distributed near intervertebral joints. These inhibitory neurons detect dynamic mechanical tension resulting from body bending, functioning as intrinsic sensors of spinal curvature[175]. **e**, The Piezo2$^+$ inner proprioceptors send contralateral, ascending projections that selectively inhibit rhythm-generating V2a interneurons (green). This circuit, therefore, integrates sensory input directly into the rhythm-generating core of the locomotor circuit, forming a hybrid sensorimotor CPG. By terminating V2a activity, these proprioceptors modulate motor neuron output (yellow), thereby aligning central motor drive with real-time body movement. Parts **c**–**e** are adapted with permission from ref. 15, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

sensory processing delays were too long to enable real-time integration. Recent discoveries challenge this classical view, suggesting that some proprioceptive neurons do not merely provide inputs to the spinal cord but are located within it and, hence, physically embedded within the CPG, forming an 'intrinsic' sensory system. These centrally located proprioceptors, thus far only described in non-mammalian species, allow for immediate, phase-specific modulation of CPG activity[174,175] (Fig. 6b). In addition to these proprioceptors, the spinal cord contains other centrally embedded sensory neurons, such as cerebrospinal fluid-contacting neurons that act as multimodal chemosensors and

Glossary

Flexor–extensor alternation
The neural mechanism that coordinates movement around one joint (for example, hip, knee or ankle) by ensuring that antagonistic flexor and extensor muscle groups are activated in an alternating sequence.

Gait
A distinct pattern of limb coordination during locomotion (for example, walk, trot or gallop) characterized by a specific timing and sequence of limb movements.

Gap junctions
Specialized intercellular connections formed by specific channels that allow the direct transfer of ions and small molecules between cells, mediating rapid electrical coupling.

Left–right alternation
The coordinated pattern wherein muscles on opposite sides of the body activate in anti-phase to produce rhythmic forward movement.

Locomotor phases
The distinct components of a locomotor cycle, such as the stance phase (limb contact with ground) and swing phase (limb movement through air), or the left–right contraction and relaxation phases in swimming.

Morphogen
A signalling molecule that emanates from a specific source and spreads to form a concentration gradient, determining the fate of developing cells based on the concentration they are exposed to.

Motor programmes
Neural circuits or commands that generate specific, coordinated movement sequences.

Proprioceptors
Sensory neurons located in muscles, joints and the skin (or within the spinal cord in some species) that detect body position, movement and mechanical load.

Recurrent connections
Synaptic connections wherein a neuron projects back onto itself or onto other neurons of the same population, creating a feedback loop often used to sustain rhythmic activity.

Single-cell RNA sequencing
A technique that analyses the gene expression profiles of individual cells, allowing the classification of neurons based on their unique molecular signatures.

mechanosensors and Rohon–Beard mechanoreceptors that convey cutaneous inputs from the periphery to the spinal cord[15,176–178]. This intrinsic feedback architecture dramatically reduces delays and tightly couples sensory detection to rhythm generation.

In the lamprey, one type of centrally located proprioceptive neurons known as edge cells line the lateral margins of the spinal cord[174]. These neurons comprise ipsilaterally projecting excitatory and contralaterally projecting inhibitory types that detect spinal bending and deliver, respectively, excitatory input ipsilaterally and inhibitory input contralaterally to CPG interneurons and motor neurons, helping to entrain and shape the locomotor rhythm[179]. Similar stretch-sensitive spinal neurons have been reported in dogfish[180], suggesting an evolutionarily conserved mechanism of sensorimotor integration across aquatic vertebrates. In adult zebrafish, a centrally located proprioceptive organ has been identified at the intervertebral joints, wherein spinal tension peaks during swimming (Fig. 6c,d). It contains glycinergic, Piezo2-expressing neurons marked by expression of Dbx1, which project contralaterally to inhibit rhythm-generating V2a interneurons in a phase-specific manner[15,175] (Fig. 6d,e). This arrangement provides rapid, real-time sensory feedback that regulates the timing of the left–right phase transition during swimming. Whether a homologous centrally embedded proprioceptive system exists in the mammalian spinal cord remains unknown.

These findings mark a fundamental shift in our understanding: the locomotor CPG is not a hardwired motor engine but a hybrid spinal sensorimotor circuit that is continuously shaped by intrinsic feedback (Fig. 6b). By embedding sensory neurons within central motor circuits, central proprioceptors reconcile the long-standing debate between Sherrington and Brown, demonstrating that sensory input and motor generation are not opposing forces but deeply intertwined components of a unified sensorimotor locomotor circuit.

Conclusions and outlook

The CPG concept is a unifying principle for centrally generated motor programmes across model systems and species. From early discoveries in cats, lamprey and *Xenopus* to recent genetic and circuit-level analyses in mice and zebrafish, the CPG model has evolved in parallel with our growing appreciation of the diversity and complexity of locomotor behaviours across species. Although this Review focuses on vertebrate systems, foundational insights from invertebrates, particularly in rhythm generation and sensorimotor integration, have critically shaped the field and are reviewed in detail elsewhere[181–184].

This Review traces the conceptual evolution of the locomotor CPG. Although early frameworks emphasized a central network of interneurons responsible for generating rhythmic activity, it is now clear that such a model alone cannot fully account for the flexibility and adaptability of natural locomotion. The notion of a hierarchically organized CPG, with discrete layers producing distinct components of the locomotor rhythmic pattern, is increasingly giving way to a more modular and dynamic view. In particular, studies in adult zebrafish have provided compelling evidence that the CPG is not a single, uniform network operating in synchrony. Instead, it is composed of distinct speed modules, each recruited according to the type of muscle activated and the vigour or speed of the locomotor behaviour. This framework introduces a major conceptual shift: the CPG is not a fixed architecture but a dynamic assembly of overlapping microcircuits tailored to behavioural demands. Although now well-established in zebrafish, this modular organization is best viewed as a candidate conserved principle across vertebrates rather than a proven universal feature, given that there are a number of important open questions regarding its generality.

Two components long considered external to the CPG, motor neurons and proprioceptors, are now being integrated into its core definition. Motor neurons, once viewed as passive output relays, are now recognized as active circuit elements. In particular, work in adult zebrafish shows that they can modulate upstream premotor interneurons, including V2a neurons, via electrical coupling. This retrograde influence allows motor neurons to dynamically shape the very locomotor patterns that they transmit, establishing them as bona fide members of the CPG. Proprioceptors, once thought to provide slow, peripheral and modulatory feedback, have now been shown to include centrally embedded variants within CPG circuits. In non-mammalian species, these central proprioceptors detect spinal bending and directly modulate rhythm-generating interneurons in a phase-dependent manner. Their dual identity both as mechanosensors and inhibitory interneurons eliminates the temporal lag traditionally associated with sensory feedback. Together, these findings blur the traditional boundary between central and peripheral components, revealing that the vertebrate CPG is a hybrid sensorimotor circuit

continuously shaped by motor neurons and proprioceptive neurons embedded within its architecture.

Outstanding questions remain at the level of the spinal locomotor CPG. One major challenge is to determine how inhibitory interneuron populations sculpt rhythm generation, regulate speed-dependent alternation and contribute to gait selection, and how they are selectively activated by brainstem commands. Another important goal is to understand how axial and appendicular CPGs interact to coordinate trunk and limb movements during locomotion. A further unresolved question concerns whether mammals possess intraspinal proprioceptive neurons analogous to those discovered in fish and lamprey, and how such intrinsic sensory feedback is incorporated into spinal computations. Finally, a central objective is to establish the precise connectivity of the mammalian CPG, to determine whether it exhibits a modular, molecularly defined, speed-specific organization comparable to that of adult zebrafish, and to define how such architecture supports the biomechanics of limbed locomotion. Addressing these challenges will be essential for clarifying which aspects of CPG organization are conserved across vertebrates and which represent species-specific specializations.

It is also now clear that brainstem locomotor commands encode much more than just movement initiation and termination. These commands also convey detailed information about the duration, speed and directionality of locomotion. Although the spinal CPG executes these commands, their final form is elaborated within local brainstem circuits that interact to form task-specific behavioural ensembles. Recent technological advances have been instrumental in revealing the basic architecture of these circuits. It is now clear that distinct brainstem populations govern different aspects of locomotor control. Some define the boundaries of locomotor episodes, whereas others act as speed and trajectory controllers, together providing a neural dial for tuning locomotor parameters. Despite this progress, many critical questions remain. These include understanding the local microcircuit interactions within brainstem nuclei, as well as the long-range connectivity between different functional modules. Addressing these will be essential to uncover how brainstem ensembles integrate multimodal sensory inputs and produce coordinated motor output. Importantly, brainstem commands must be continuously updated based on the state of the spinal CPG and the body. This raises the question of how ascending information from the spinal cord is relayed to and processed within brainstem structures. Finally, the interface between descending commands and distinct CPG modules remains poorly understood. Understanding these interactions will be key to linking top–down control with bottom–up sensorimotor execution.

Looking forwards, the field is poised to move beyond static maps of circuit architecture towards dynamic models that incorporate real-time neuromodulation, biomechanics and sensory feedback. A central challenge will be to bridge scales, from ion channels, synaptic and circuit motifs, to whole-animal behaviour, and to understand how CPG circuits are flexibly engaged and reconfigured to support diverse motor tasks. Ultimately, such integrative approaches will be essential to uncover the general principles by which neural circuits produce movement that is both robust and adaptive across the animal kingdom.

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References

1. Grillner, S. *The Brain in Motion: From Microcircuits to Global Brain Function* (The MIT Press, 2023).
2. Orlovsky, G. N., Deliagina, T. G. & Grillner, S. *Neuronal Control of Locomotion: From Mollusc to Man* (Oxford Univ. Press, 1999).
3. Sillar, K. T., Picton, L. D. & Heitler, W. J. *The Neuroethology of Predation and Escape* (Wiley, 2016).
4. Berg, E. M., Bjornfors, E. R., Pallucchi, I., Picton, L. D. & El Manira, A. Principles governing locomotion in vertebrates: lessons from zebrafish. *Front. Neural Circuits* **12**, 73 (2018).
5. Brownstone, R. M. & Bui, T. V. Spinal interneurons providing input to the final common path during locomotion. *Prog. Brain Res.* **187**, 81–95 (2010).
6. Büschges, A., Scholz, H. & El Manira, A. New moves in motor control. *Curr. Biol.* **21**, R513–R524 (2011).
7. Dougherty, K. J. Distinguishing subtypes of spinal locomotor neurons to inform circuit function and dysfunction. *Curr. Opin. Neurobiol.* **82**, 102763 (2023).
8. Gosgnach, S. et al. Delineating the diversity of spinal interneurons in locomotor circuits. *J. Neurosci.* **37**, 10835–10841 (2017).
9. Goulding, M. Circuits controlling vertebrate locomotion: moving in a new direction. *Nat. Rev. Neurosci.* **10**, 507–518 (2009).
10. Grillner, S. Locomotion in vertebrates: central mechanisms and reflex interaction. *Physiol. Rev.* **55**, 247–304 (1975).
11. Grillner, S. The motor infrastructure: from ion channels to neuronal networks. *Nat. Rev. Neurosci.* **4**, 573–586 (2003).
12. Grillner, S. & El Manira, A. Current principles of motor control, with special reference to vertebrate locomotion. *Physiol. Rev.* **100**, 271–320 (2020).
A comprehensive synthesis of the physiological principles governing vertebrate locomotion; this review compares circuit architectures across species from lamprey to mammals, highlighting the conserved organizational principles.
13. Kiehn, O. Decoding the organization of spinal circuits that control locomotion. *Nat. Rev. Neurosci.* **17**, 224–238 (2016).
A seminal review outlining the genetic logic of the mammalian spinal CPG; it details how transcriptionally defined interneuron classes (V0–V3) map onto specific functional roles in rhythm generation, left–right alternation and speed regulation.
14. McLean, D. L. & Dougherty, K. J. Peeling back the layers of locomotor control in the spinal cord. *Curr. Opin. Neurobiol.* **33**, 63–70 (2015).
15. Picton, L. et al. Circuit modules for flexible locomotion. *Annu. Rev. Neurosci.* **48**, 23–41 (2025).
16. Roberts, A., Li, W. C. & Soffe, S. R. How neurons generate behavior in a hatchling amphibian tadpole: an outline. *Front. Behav. Neurosci.* **4**, 16 (2010).
17. Ryczko, D., Simon, A. & Ijspeert, A. J. Walking with salamanders: from molecules to biorobotics. *Trends Neurosci.* **43**, 916–930 (2020).
18. Sengupta, M. & Bagnall, M. W. Spinal interneurons: diversity and connectivity in motor control. *Annu. Rev. Neurosci.* **46**, 79–99 (2023).
19. Sillar, K. T., Simmers, J. & Combes, D. From tadpole to adult frog locomotion. *Curr. Opin. Neurobiol.* **82**, 102753 (2023).
20. Arber, S. & Costa, R. M. Networking brainstem and basal ganglia circuits for movement. *Nat. Rev. Neurosci.* **23**, 342–360 (2022).
Establishes a seminal framework for the functional hierarchy between the basal ganglia and brainstem in motor control; the authors delineate how basal ganglia circuits, which select actions, interface while dedicating brainstem modules that define the specific parameters of movement, such as speed, direction and posture.
21. Brownstone, R. M. & Chopek, J. W. Reticulospinal systems for tuning motor commands. *Front. Neural Circuits* **12**, 30 (2018).
22. Jordan, L. M. Initiation of locomotion in mammals. *Ann. N. Y. Acad. Sci.* **860**, 83–93 (1998).
23. Leiras, R., Cregg, J. M. & Kiehn, O. Brainstem circuits for locomotion. *Annu. Rev. Neurosci.* **45**, 63–85 (2022).
24. Ryczko, D. The mesencephalic locomotor region: multiple cell types, multiple behavioral roles, and multiple implications for disease. *Neuroscientist* **30**, 347–366 (2024).
25. Ryczko, D. & Dubuc, R. The multifunctional mesencephalic locomotor region. *Curr. Pharm. Des.* **19**, 4448–4470 (2013).
26. Thiele, T. R., Donovan, J. C. & Baier, H. Descending control of swim posture by a midbrain nucleus in zebrafish. *Neuron* **83**, 679–691 (2014).
Demonstrates that the nMLF controls swim posture and steering in addition to speed in larval zebrafish; the study identifies specific neurons that control the gain of the vestibulo-spinal reflex to maintain balance during locomotion.
27. Baier, H. & Scott, E. K. The visual systems of zebrafish. *Annu. Rev. Neurosci.* **47**, 255–276 (2024).
28. Roome, R. B. & Levine, A. J. The organization of spinal neurons: insights from single cell sequencing. *Curr. Opin. Neurobiol.* **82**, 102762 (2023).
29. Sagner, A. & Briscoe, J. Establishing neuronal diversity in the spinal cord: a time and a place. *Development* **146**, dev182154 (2019).
30. Tuthill, J. C. & Azim, E. Proprioception. *Curr. Biol.* **28**, R194–R203 (2018).
A broad overview of the mechanisms and computational roles of proprioception; it contrasts the classical view of feedback as a reflex correction system with recent evidence that proprioceptive circuits are integral to the planning and ongoing regulation of complex motor behaviours.
31. Wilson, A. C. & Sweeney, L. B. Spinal cords: symphonies of interneurons across species. *Front. Neural Circuits* **17**, 1146449 (2023).
32. Caggiano, V. et al. Midbrain circuits that set locomotor speed and gait selection. *Nature* **553**, 455–460 (2018).
Dissects the mouse MLR into two functional dials; the authors show that the CnF drives high-speed escape locomotion, whereas the PPN regulates slow, exploratory movement, linking anatomical subregions to behavioural context.

33. Capelli, P., Pivetta, C., Soledad Esposito, M. & Arber, S. Locomotor speed control circuits in the caudal brainstem. *Nature* **551**, 373–377 (2017).
Identifies specific populations in the medullary reticular formation (LPGi) that mediate the locomotor commands from the MLR; the study also characterizes a separate population of 'stop neurons' in the brainstem that can override locomotor signals to halt movement.
34. Carbo-Tano, M. et al. The mesencephalic locomotor region recruits V2a reticulospinal neurons to drive forward locomotion in larval zebrafish. *Nat. Neurosci.* **26**, 1775–1790 (2023).
Maps the descending pathway from the MLR in zebrafish, demonstrating that it executes forward swimming by recruiting specific V2a reticulospinal neurons.
35. Carvalho, M. M. et al. A brainstem locomotor circuit drives the activity of speed cells in the medial entorhinal cortex. *Cell Rep.* **32**, 108123 (2020).
36. El Manira, A., Pombal, M. A. & Grillner, S. Diencephalic projection to reticulospinal neurons involved in the initiation of locomotion in adult lampreys *Lampetra fluviatilis*. *J. Comp. Neurol.* **389**, 603–616 (1997).
37. Josset, N. et al. Distinct contributions of mesencephalic locomotor region nuclei to locomotor control in the freely behaving mouse. *Curr. Biol.* **28**, 884–901.e3 (2018).
Investigates the MLR during voluntary movement, showing that the CnF and PPN have distinct activation profiles; the CnF is preferentially recruited during high-speed surges, whereas the PPN is active during slower, maintenance locomotion, supporting the functional division of the MLR.
38. Shik, M. L. & Orlovsky, G. N. Neurophysiology of locomotor automatism. *Physiol. Rev.* **56**, 465–501 (1976).
39. Ausborn, J., Mahmood, R. & El Manira, A. Decoding the rules of recruitment of excitatory interneurons in the adult zebrafish locomotor network. *Proc. Natl Acad. Sci. USA* **109**, E3631–E3639 (2012).
40. Ferreira-Pinto, M. J. et al. Functional diversity for body actions in the mesencephalic locomotor region. *Cell* **184**, 4564–4578.e18 (2021).
Reveals that the MLR controls not just locomotion but a repertoire of body actions, including rearing and handling; the authors show that specific cell types within the MLR encode these distinct motor programmes, broadening the view of the MLR.
41. Roseberry, T. K. et al. Cell-type-specific control of brainstem locomotor circuits by basal ganglia. *Cell* **164**, 526–537 (2016).
42. Kimura, Y. et al. Hindbrain V2a neurons in the excitation of spinal locomotor circuits during zebrafish swimming. *Curr. Biol.* **23**, 843–849 (2013).
43. Berg, E. M. et al. Brainstem circuits encoding start, speed, and duration of swimming in adult zebrafish. *Neuron* **111**, 372–386.e4 (2023).
Identifies the cellular substrate of the dual-dial model in the zebrafish nMLF; the study reveals that Vglut2$^+$ neurons encode the initiation and duration of swimming, whereas a distinct Vglut1$^+$ population encodes swimming speed, allowing for combinatorial control of locomotor episodes.
44. Severi, K. E. et al. Neural control and modulation of swimming speed in the larval zebrafish. *Neuron* **83**, 692–707 (2014).
Establishes a functional map of the larval zebrafish brainstem, showing that neurons are organized topographically according to the swimming speeds they control.
45. Wang, W. C. & McLean, D. L. Selective responses to tonic descending commands by temporal summation in a spinal motor pool. *Neuron* **83**, 708–721 (2014).
46. El Manira, A. Modular circuit organization for speed control of locomotor movements. *Curr. Opin. Neurobiol.* **82**, 102760 (2023).
47. Dautan, D. et al. Modulation of motor behavior by the mesencephalic locomotor region. *Cell Rep.* **36**, 109594 (2021).
Explores how the MLR modulates motor behaviour in real-time, focusing on the integration of sensory and internal state signals to adjust locomotor intensity.
48. Armstrong, D. M. Supraspinal contributions to the initiation and control of locomotion in the cat. *Prog. Neurobiol.* **26**, 273–361 (1986).
49. Drew, T., Fortier-Lebel, N. & Nakajima, T. Cortical contribution to visuomotor coordination in locomotion and reaching. *Curr. Opin. Neurobiol.* **82**, 102755 (2023).
50. Drew, T. & Marigold, D. S. Taking the next step: cortical contributions to the control of locomotion. *Curr. Opin. Neurobiol.* **33**, 25–33 (2015).
51. Bouvier, J. et al. Descending command neurons in the brainstem that halt locomotion. *Cell* **163**, 1191–1203 (2015).
Identifies a population of reticulospinal neurons (CHX10$^+$) in the brainstem that are dedicated to stopping locomotion; activation of these neurons arrests the spinal CPG, showing that 'stop' is an active command rather than just the absence of a 'go' signal.
52. Schwenkgrub, J., Harrell, E. R., Bathellier, B. & Bouvier, J. Deep imaging in the brainstem reveals functional heterogeneity in V2a neurons controlling locomotion. *Sci. Adv.* **6**, eabc6309 (2020).
53. Usseglio, G., Gatier, E., Heuzé, A., Hérent, C. & Bouvier, J. Control of orienting movements and locomotion by projection-defined subsets of brainstem V2a neurons. *Curr. Biol.* **30**, 4665–4681.e6 (2020).
Dissects the V2a reticulospinal population in mice, showing that distinct subsets control forward locomotion versus orienting movements; this highlights the modular organization of the descending reticulospinal system for different motor synergies.
54. Cregg, J. M. et al. Brainstem neurons that command mammalian locomotor asymmetries. *Nat. Neurosci.* **23**, 730–740 (2020).
Identifies CHX10$^+$ reticulospinal neurons as the command centre for turning in mice; asymmetric activation of these neurons biases spinal circuits to produce left–right asymmetries, providing a defined mechanism for steering.
55. Cregg, J. M., Sidhu, S. K., Leiras, R. & Kiehn, O. Basal ganglia-spinal cord pathway that commands locomotor gait asymmetries in mice. *Nat. Neurosci.* **27**, 716–727 (2024).
56. Huang, K. H., Ahrens, M. B., Dunn, T. W. & Engert, F. Spinal projection neurons control turning behaviors in zebrafish. *Curr. Biol.* **23**, 1566–1573 (2013).
Shows that routine turning in zebrafish is controlled by specific descending neurons in the hindbrain, independent of the Mauthner escape system; these neurons project asymmetrically to the spinal cord to drive directional swimming.
57. Xu, L. et al. Separate brainstem circuits for fast steering and slow exploratory turns. *Nat. Commun.* **16**, 3207 (2025).
Demonstrates that steering control is segregated into speed-dependent channels; the study identifies distinct brainstem circuits for executing slow, exploratory turns versus fast, steering turns, extending the 'speed module' concept to directional control.
58. Sherrington, C. S. Flexion-reflex of the limb, crossed extension-reflex, and reflex stepping and standing. *J. Physiol.* **40**, 28–121 (1910).
59. Brown, T. G. The intrinsic factors in the act of progression in the mammal. *Proc. R. Soc. Lond. B* **84**, 308–319 (1911).
60. Lundberg, A. *Nansen Memorial Lecture V* (Universitetsforlaget, 1969).
61. Grillner, S. & Zangger, P. How detailed is the central pattern generation for locomotion? *Brain Res.* **88**, 367–371 (1975).
62. Grillner, S. & Zangger, P. The effect of dorsal root transection on the efferent motor pattern in the cat's hindlimb during locomotion. *Acta Physiol. Scand.* **120**, 393–405 (1984).
63. Akay, T. Sensory feedback control of locomotor pattern generation in cats and mice. *Neuroscience* **450**, 161–167 (2020).
64. Pearson, K. G. Role of sensory feedback in the control of stance duration in walking cats. *Brain Res. Rev.* **57**, 222–227 (2008).
65. Rossignol, S., Dubuc, R. & Gossard, J. P. Dynamic sensorimotor interactions in locomotion. *Physiol. Rev.* **86**, 89–154 (2006).
66. Arshavsky Yu, I., Orlovsky, G. N., Panchin Yu, V., Roberts, A. & Soffe, S. R. Neuronal control of swimming locomotion: analysis of the Pteropod mollusc *Clione* and embryos of the amphibian *Xenopus*. *Trends Neurosci.* **16**, 227–233 (1993).
67. Buchanan, J. T. The roles of spinal interneurons and motoneurons in the lamprey locomotor network. *Prog. Brain Res.* **123**, 311–321 (1999).
68. Buchanan, J. T. Contributions of identifiable neurons and neuron classes to lamprey vertebrate neurobiology. *Prog. Neurobiol.* **63**, 441–466 (2001).
69. Grillner, S. Biological pattern generation: the cellular and computational logic of networks in motion. *Neuron* **52**, 751–766 (2006).
70. Grillner, S. & Jessell, T. M. Measured motion: searching for simplicity in spinal locomotor networks. *Curr. Opin. Neurobiol.* **19**, 572–586 (2009).
71. Hachoumi, L. & Sillar, K. T. Developmental stage-dependent switching in the neuromodulation of vertebrate locomotor central pattern generator networks. *Dev. Neurobiol.* **80**, 42–57 (2020).
72. McLean, D. L., Merrywest, S. D. & Sillar, K. T. The development of neuromodulatory systems and the maturation of motor patterns in amphibian tadpoles. *Brain Res. Bull.* **53**, 595–603 (2000).
73. Roberts, A. et al. The decision to move: response times, neuronal circuits and sensory memory in a simple vertebrate. *Proc. Biol. Sci.* **286**, 20190297 (2019).
74. Roberts, A., Li, W. C. & Soffe, S. R. A functional scaffold of CNS neurons for the vertebrates: the developing *Xenopus laevis* spinal cord. *Dev. Neurobiol.* **72**, 575–584 (2012).
75. Dasen, J. S. Transcriptional networks in the early development of sensory-motor circuits. *Curr. Top. Dev. Biol.* **87**, 119–148 (2009).
76. Goulding, M. & Pfaff, S. L. Development of circuits that generate simple rhythmic behaviors in vertebrates. *Curr. Opin. Neurobiol.* **15**, 14–20 (2005).
77. Jessell, T. M. Neuronal specification in the spinal cord: inductive signals and transcriptional codes. *Nat. Rev. Genet.* **1**, 20–29 (2000).
78. Osseward, P. J. II et al. Conserved genetic signatures parcellate cardinal spinal neuron classes into local and projection subsets. *Science* **372**, 385–393 (2021).
79. D'Elia, K. P. & Dasen, J. S. Development, functional organization, and evolution of vertebrate axial motor circuits. *Neural Dev.* **13**, 10 (2018).
80. Osseward, P. J. II & Pfaff, S. L. Cell type and circuit modules in the spinal cord. *Curr. Opin. Neurobiol.* **56**, 175–184 (2019).
81. Arber, S. Motor circuits in action: specification, connectivity, and function. *Neuron* **74**, 975–989 (2012).
82. Ampatzis, K., Song, J., Ausborn, J. & El Manira, A. Separate microcircuit modules of distinct v2a interneurons and motoneurons control the speed of locomotion. *Neuron* **83**, 934–943 (2014).
Establishes the modular architecture of the zebrafish CPG, showing that it is composed of three discrete speed-tuned circuits (slow, intermediate, fast); specific V2a interneurons connect exclusively to motor neurons of the same speed class, ensuring precise recruitment.
83. Eklof-Ljunggren, E. et al. Origin of excitation underlying locomotion in the spinal circuit of zebrafish. *Proc. Natl Acad. Sci. USA* **109**, 5511–5516 (2012).
84. Ljunggren, E. E., Haupt, S., Ausborn, J., Ampatzis, K. & El Manira, A. Optogenetic activation of excitatory premotor interneurons is sufficient to generate coordinated locomotor activity in larval zebrafish. *J. Neurosci.* **34**, 134–139 (2014).
85. Song, J. et al. Multiple rhythm-generating circuits act in tandem with pacemaker properties to control the start and speed of locomotion. *Neuron* **105**, 1048–1061.e4 (2020).
Reveals the synaptic hierarchy between speed modules; slow modules possess strong pacemaker properties to initiate rhythm, whereas fast modules provide top–down, unidirectional drive to slower circuits to coordinate high-speed swimming.

86. Chalif, J. I., Martínez-Silva, M. L., Pagiazitis, J. G., Murray, A. J. & Mentis, G. Z. Control of mammalian locomotion by ventral spinocerebellar tract neurons. *Cell* **185**, 328–344.e6 (2022).
87. Dougherty, K. J. et al. Locomotor rhythm generation linked to the output of spinal Shox2 excitatory interneurons. *Neuron* **80**, 920–933 (2013).
88. Ha, N. T. & Dougherty, K. J. Spinal Shox2 interneuron interconnectivity related to function and development. *eLife* **7**, e42519 (2018).
89. Crone, S. A. et al. Genetic ablation of V2a ipsilateral interneurons disrupts left-right locomotor coordination in mammalian spinal cord. *Neuron* **60**, 70–83 (2008).
90. Zhong, G., Sharma, K. & Harris-Warrick, R. M. Frequency-dependent recruitment of V2a interneurons during fictive locomotion in the mouse spinal cord. *Nat. Commun.* **2**, 274 (2011).
Demonstrates that V2a interneurons in mice are recruited in a frequency-dependent manner; this suggests that the principle of speed-dependent recruitment is conserved between zebrafish and mammals.
91. Kwan, A. C., Dietz, S. B., Webb, W. W. & Harris-Warrick, R. M. Activity of Hb9 interneurons during fictive locomotion in mouse spinal cord. *J. Neurosci.* **29**, 11601–11613 (2009).
92. Koronfel, L. M., Kanning, K. C., Alcos, A., Henderson, C. E. & Brownstone, R. M. Elimination of glutamatergic transmission from Hb9 interneurons does not impact treadmill locomotion. *Sci. Rep.* **11**, 16008 (2021).
93. Bertho, M. et al. Excitatory spinal Lhx9-derived interneurons modulate locomotor frequency in mice. *J. Neurosci.* **44**, e1607232024 (2024).
94. Crone, S. A., Zhong, G., Harris-Warrick, R. & Sharma, K. In mice lacking V2a interneurons, gait depends on speed of locomotion. *J. Neurosci.* **29**, 7098–7109 (2009).
95. Brocard, F., Tazerart, S. & Vinay, L. Do pacemakers drive the central pattern generator for locomotion in mammals? *Neuroscientist* **16**, 139–155 (2010).
96. Kimura, Y., Okamura, Y. & Higashijima, S. alx, a zebrafish homolog of Chx10, marks ipsilateral descending excitatory interneurons that participate in the regulation of spinal locomotor circuits. *J. Neurosci.* **26**, 5684–5697 (2006).
The initial characterization of V2a (Alx/Chx10) neurons in zebrafish; this study has established their role as the primary source of excitatory drive to motor neurons and their necessity for robust swimming activity.
97. Song, J., Dahlberg, E. & El Manira, A. V2a interneuron diversity tailors spinal circuit organization to control the vigor of locomotor movements. *Nat. Commun.* **9**, 3370 (2018).
98. Brocard, F. et al. Activity-dependent changes in extracellular Ca^{2+} and K^{+} reveal pacemakers in the spinal locomotor-related network. *Neuron* **77**, 1047–1054 (2013).
99. Tazerart, S., Vinay, L. & Brocard, F. The persistent sodium current generates pacemaker activities in the central pattern generator for locomotion and regulates the locomotor rhythm. *J. Neurosci.* **28**, 8577–8589 (2008).
100. Wilson, J. M., Cowan, A. I. & Brownstone, R. M. Heterogeneous electrotonic coupling and synchronization of rhythmic bursting activity in mouse Hb9 interneurons. *J. Neurophysiol.* **98**, 2370–2381 (2007).
101. Gabriel, J. P. et al. Principles governing recruitment of motoneurons during swimming in zebrafish. *Nat. Neurosci.* **14**, 93–99 (2011).
102. Ampatzis, K., Song, J., Ausborn, J. & El Manira, A. Pattern of innervation and recruitment of different classes of motoneurons in adult zebrafish. *J. Neurosci.* **33**, 10875–10886 (2013).
103. Burke, R. E. in *Handbook of Physiology*, *Section 1*: *The Nervous System* Vol. 1, 345–422 (American Physiological Society, 1981).
104. Dasen, J. S. & Jessell, T. M. Hox networks and the origins of motor neuron diversity. *Curr. Top. Dev. Biol.* **88**, 169–200 (2009).
105. Henneman, E. & Mendell, L. M. in *Handbook of Physiology, Section 1*: *The Nervous System* Vol. 2 (ed. Brooks, V. E.) 423–507 (American Physiological Society, 1981).
106. Henneman, E. Relation between size of neurons and their susceptibility to discharge. *Science* **126**, 1345–1347 (1957).
107. Mendell, L. M. The size principle: a rule describing the recruitment of motoneurons. *J. Neurophysiol.* **93**, 3024–3026 (2005).
108. McLean, D. L., Fan, J., Higashijima, S., Hale, M. E. & Fetcho, J. R. A topographic map of recruitment in spinal cord. *Nature* **446**, 71–75 (2007).
A foundational study showing that spinal neurons in zebrafish are recruited in a ventral-to-dorsal topographic sequence as speed increases; this links the physical position of a neuron in the spinal cord directly to its functional speed tuning.
109. Bikoff, J. B. et al. Spinal inhibitory interneuron diversity delineates variant motor microcircuits. *Cell* **165**, 207–219 (2016).
110. Gabitto, M. I. et al. Bayesian sparse regression analysis documents the diversity of spinal inhibitory interneurons. *Cell* **165**, 220–233 (2016).
111. Alkaslasi, M. R. et al. Single nucleus RNA-sequencing defines unexpected diversity of cholinergic neuron types in the adult mouse spinal cord. *Nat. Commun.* **12**, 2471 (2021).
112. Blum, J. A. et al. Single-cell transcriptomic analysis of the adult mouse spinal cord reveals molecular diversity of autonomic and skeletal motor neurons. *Nat. Neurosci.* **24**, 572–583 (2021).
113. D'Elia, K. P. et al. Determinants of motor neuron functional subtypes important for locomotor speed. *Cell Rep.* **42**, 113049 (2023).
114. Delile, J. et al. Single cell transcriptomics reveals spatial and temporal dynamics of gene expression in the developing mouse spinal cord. *Development* **146**, dev173807 (2019).
115. Rayon, T., Maizels, R. J., Barrington, C. & Briscoe, J. Single-cell transcriptome profiling of the human developing spinal cord reveals a conserved genetic programme with human-specific features. *Development* **148**, dev199711 (2021).
116. Russ, D. E. et al. A harmonized atlas of mouse spinal cord cell types and their spatial organization. *Nat. Commun.* **12**, 5722 (2021).
117. Yadav, A. et al. A cellular taxonomy of the adult human spinal cord. *Neuron* **111**, 328–344.e7 (2023).
118. Pallucchi, I. et al. Molecular blueprints for spinal circuit modules controlling locomotor speed in zebrafish. *Nat. Neurosci.* **27**, 78–89 (2024).
Provides the transcriptional signatures for the speed-dependent modules in adult zebrafish; the authors identify specific genetic markers that distinguish slow, intermediate and fast V2a interneurons and motor neurons.
119. Kelly, J. J., Wen, H. & Brehm, P. Single-cell RNAseq analysis of spinal locomotor circuitry in larval zebrafish. *eLife* **12**, e89338 (2023).
120. Goulding, M., Bourane, S., Garcia-Campmany, L., Dalet, A. & Koch, S. Inhibition downunder: an update from the spinal cord. *Curr. Opin. Neurobiol.* **26**, 161–166 (2014).
121. Moult, P. R., Cottrell, G. A. & Li, W. C. Fast silencing reveals a lost role for reciprocal inhibition in locomotion. *Neuron* **77**, 129–140 (2013).
Re-evaluates the role of inhibitory pathways in rhythm generation; the authors demonstrate in *Xenopus* tadpoles that reciprocal inhibition is essential for maintaining the locomotor rhythm, challenging models that rely solely on excitatory pacemakers.
122. Cohen, A. H. & Harris-Warrick, R. M. Strychnine eliminates alternating motor output during fictive locomotion in the lamprey. *Brain Res.* **293**, 164–167 (1984).
123. Cowley, K. C. & Schmidt, B. J. Regional distribution of the locomotor pattern-generating network in the neonatal rat spinal cord. *J. Neurophysiol.* **77**, 247–259 (1997).
124. Gabriel, J. P. et al. Serotonergic modulation of locomotion in zebrafish: endogenous release and synaptic mechanisms. *J. Neurosci.* **29**, 10387–10395 (2009).
125. Gabriel, J. P. et al. Locomotor pattern in the adult zebrafish spinal cord in vitro. *J. Neurophysiol.* **99**, 37–48 (2008).
126. Pierani, A. et al. Control of interneuron fate in the developing spinal cord by the progenitor homeodomain protein Dbx1. *Neuron* **29**, 367–384 (2001).
127. Lanuza, G. M., Gosgnach, S., Pierani, A., Jessell, T. M. & Goulding, M. Genetic identification of spinal interneurons that coordinate left-right locomotor activity necessary for walking movements. *Neuron* **42**, 375–386 (2004).
Identifies V0 commissural interneurons as the essential component for left–right alternation in mice.
128. Talpalar, A. E. et al. Dual-mode operation of neuronal networks involved in left-right alternation. *Nature* **500**, 85–88 (2013).
Shows that left–right coordination is handled by distinct V0 subcircuits depending on speed; inhibitory V0d neurons control slow walking, whereas excitatory V0v neurons are recruited at high speeds to maintain alternation via inhibitory intermediates.
129. Bellardita, C. & Kiehn, O. Phenotypic characterization of speed-associated gait changes in mice reveals modular organization of locomotor networks. *Curr. Biol.* **25**, 1426–1436 (2015).
130. Zelenin, P. V. et al. Differential contribution of V0 interneurons to execution of rhythmic and nonrhythmic motor behaviors. *J. Neurosci.* **41**, 3432–3445 (2021).
131. Bjornfors, E. R. & El Manira, A. Functional diversity of excitatory commissural interneurons in adult zebrafish. *eLife* **5**, e18579 (2016).
132. Picton, L. D. et al. Developmental switch in the function of inhibitory commissural V0d interneurons in zebrafish. *Curr. Biol.* **32**, 3515–3528.e4 (2022).
133. Satou, C., Kimura, Y. & Higashijima, S. Generation of multiple classes of V0 neurons in zebrafish spinal cord: progenitor heterogeneity and temporal control of neuronal diversity. *J. Neurosci.* **32**, 1771–1783 (2012).
134. Satou, C. et al. Functional diversity of glycinergic commissural inhibitory neurons in larval zebrafish. *Cell Rep.* **30**, 3036–3050.e4 (2020).
135. Kishore, S., Cadoff, E. B., Agha, M. A. & McLean, D. L. Orderly compartmental mapping of premotor inhibition in the developing zebrafish spinal cord. *Science* **370**, 431–436 (2020).
136. Fidelin, K. et al. State-dependent modulation of locomotion by GABAergic spinal sensory neurons. *Curr. Biol.* **25**, 3035–3047 (2015).
137. Kawano, K. et al. Long descending commissural V0v neurons ensure coordinated swimming movements along the body axis in larval zebrafish. *Sci. Rep.* **12**, 4348 (2022).
138. Del Pozo, A. et al. Behavioral characterization of dmrt3a mutant zebrafish reveals crucial aspects of vertebrate locomotion through phenotypes related to acceleration. *eNeuro* **7**, ENEURO.0047-20.2020 (2020).
139. Andersson, L. S. et al. Mutations in DMRT3 affect locomotion in horses and spinal circuit function in mice. *Nature* **488**, 642–646 (2012).
Identifies a mutation in the *DMRT3* gene (marking dI6 interneurons) as the cause of 'gaited' locomotion in horses; this study links a specific spinal interneuron population to the ability to maintain specific gait patterns, such as the pace or *tölt*.
140. Perry, S. et al. Characterization of Dmrt3-derived neurons suggest a role within locomotor circuits. *J. Neurosci.* **39**, 1771–1782 (2019).
141. Böhm, U. L. et al. Voltage imaging identifies spinal circuits that modulate locomotor adaptation in zebrafish. *Neuron* **110**, 1211–1222.e4 (2022).
142. Wiggin, T. D., Montgomery, J. E., Brunick, A. J., Peck, J. H. & Masino, M. A. V3 interneurons are active and recruit spinal motor neurons during in vivo fictive swimming in larval zebrafish. *eNeuro* **9**, ENEURO.0476-21.2022 (2022).
143. Zhang, H. et al. Widespread innervation of motoneurons by spinal V3 neurons globally amplifies locomotor output in mice. *Cell Rep.* **44**, 115212 (2025).
144. Zhang, Y. et al. V3 spinal neurons establish a robust and balanced locomotor rhythm during walking. *Neuron* **60**, 84–96 (2008).
145. Jankowska, E. Spinal interneuronal systems: identification, multifunctional character and reconfigurations in mammals. *J. Physiol.* **533**, 31–40 (2001).

146. Jankowska, E. Spinal interneuronal networks in the cat: elementary components. *Brain Res. Rev.* **57**, 46–55 (2008).
147. Gosgnach, S. et al. V1 spinal neurons regulate the speed of vertebrate locomotor outputs. *Nature* **440**, 215–219 (2006).
Identifies V1 inhibitory interneurons as critical regulators of locomotor speed; loss of V1 neurons results in a drastic slowing of the locomotor rhythm, indicating that they control the cycle period of the CPG.
148. Higashijima, S., Masino, M. A., Mandel, G. & Fetcho, J. R. Engrailed-1 expression marks a primitive class of inhibitory spinal interneuron. *J. Neurosci.* **24**, 5827–5839 (2004).
149. Kimura, Y. & Higashijima, S. I. Regulation of locomotor speed and selection of active sets of neurons by V1 neurons. *Nat. Commun.* **10**, 2268 (2019).
Demonstrates that V1 interneurons in zebrafish are crucial not only for locomotor burst termination but also for selection of the active sets of neurons; this ensures that only the appropriate set of neurons is active for a given speed, sharpening the precision of the motor output.
150. Li, W. C., Higashijima, S., Parry, D. M., Roberts, A. & Soffe, S. R. Primitive roles for inhibitory interneurons in developing frog spinal cord. *J. Neurosci.* **24**, 5840–5848 (2004).
151. Sengupta, M., Daliparthi, V., Roussel, Y., Bui, T. V. & Bagnall, M. W. Spinal V1 neurons inhibit motor targets locally and sensory targets distally. *Curr. Biol.* **31**, 3820–3833.e4 (2021).
152. Zhang, J. et al. V1 and V2b interneurons secure the alternating flexor-extensor motor activity mice require for limbed locomotion. *Neuron* **82**, 138–150 (2014).
153. Callahan, R. A. et al. Spinal V2b neurons reveal a role for ipsilateral inhibition in speed control. *eLife* **8**, e47837 (2019).
154. Sengupta, M., Bertram, A., Zhu, S. I., Goodhill, G. J. & Bagnall, M. W. V2b neurons act via multiple targets to produce in phase inhibition during locomotion. *J. Neurosci.* **45**, e1530242025 (2025).
155. Wenner, P. & O'Donovan, M. J. Mechanisms that initiate spontaneous network activity in the developing chick spinal cord. *J. Neurophysiol.* **86**, 1481–1498 (2001).
156. Hanson, M. G. & Landmesser, L. T. Characterization of the circuits that generate spontaneous episodes of activity in the early embryonic mouse spinal cord. *J. Neurosci.* **23**, 587–600 (2003).
157. Bonnot, A., Chub, N., Pujala, A. & O'Donovan, M. J. Excitatory actions of ventral root stimulation during network activity generated by the disinhibited neonatal mouse spinal cord. *J. Neurophysiol.* **101**, 2995–3011 (2009).
158. Machacek, D. W. & Hochman, S. Noradrenaline unmasks novel self-reinforcing motor circuits within the mammalian spinal cord. *J. Neurosci.* **26**, 5920–5928 (2006).
159. Falgairolle, M., Puhl, J. G., Pujala, A., Liu, W. & O'Donovan, M. J. Motoneurons regulate the central pattern generator during drug-induced locomotor-like activity in the neonatal mouse. *eLife* **6**, e26622 (2017).
Shows evidence that motor neurons in mammals provide retrograde feedback to the CPG; optogenetic manipulation of motor neurons alters the frequency of the locomotor rhythm, showing that they are active participants in rhythm generation.
160. Falgairolle, M. & O'Donovan, M. J. Feedback regulation of locomotion by motoneurons in the vertebrate spinal cord. *Curr. Opin. Physiol.* **8**, 50–55 (2019).
161. Pallucchi, I., Bertuzzi, M., Michel, J. C., Miller, A. C. & El Manira, A. Transformation of an early-established motor circuit during maturation in zebrafish. *Cell Rep.* **39**, 110654 (2022).
162. Song, J., Ampatzis, K., Bjornfors, E. R. & El Manira, A. Motor neurons control locomotor circuit function retrogradely via gap junctions. *Nature* **529**, 399–402 (2016).
Shows that in zebrafish, motor neurons regulate the excitability of premotor V2a interneurons via gap junctions; this retrograde mechanism allows motor neurons to gate the recruitment of the circuits that drive them.
163. Bautista, W., McCrea, D. A. & Nagy, J. I. Connexin36 identified at morphologically mixed chemical/electrical synapses on trigeminal motoneurons and at primary afferent terminals on spinal cord neurons in adult mouse and rat. *Neuroscience* **263**, 159–180 (2014).
164. Rash, J. E. et al. Mixed synapses discovered and mapped throughout mammalian spinal cord. *Proc. Natl Acad. Sci. USA* **93**, 4235–4239 (1996).
165. Coste, B. et al. Piezo1 and Piezo2 are essential components of distinct mechanically activated cation channels. *Science* **330**, 55–60 (2010).
166. Woo, S. H. et al. Piezo2 is the principal mechanotransduction channel for proprioception. *Nat. Neurosci.* **18**, 1756–1762 (2015).
167. Woo, S. H. et al. Piezo2 is required for Merkel-cell mechanotransduction. *Nature* **509**, 622–626 (2014).
168. Nagel, M. & Chesler, A. T. PIEZO2 ion channels in proprioception. *Curr. Opin. Neurobiol.* **75**, 102572 (2022).
169. Szczot, M., Nickolls, A. R., Lam, R. M. & Chesler, A. T. The form and function of PIEZO2. *Annu. Rev. Biochem.* **90**, 507–534 (2021).
170. Chesler, A. T. et al. The role of PIEZO2 in human mechanosensation. *N. Engl. J. Med.* **375**, 1355–1364 (2016).
171. Akay, T., Tourtellotte, W. G., Arber, S. & Jessell, T. M. Degradation of mouse locomotor pattern in the absence of proprioceptive sensory feedback. *Proc. Natl Acad. Sci. USA* **111**, 16877–16882 (2014).
172. Goulding, M., Bollu, T. & Büschges, A. Sensory feedback and the dynamic control of movement. *Annu. Rev. Neurosci.* **48**, 405–423 (2025).
173. Santuz, A. & Akay, T. Muscle spindles and their role in maintaining robust locomotion. *J. Physiol.* **601**, 275–285 (2023).
174. Grillner, S., Williams, T. & Lagerbäck, P. A. The edge cell, a possible intraspinal mechanoreceptor. *Science* **223**, 500–503 (1984).
175. Picton, L. D. et al. A spinal organ of proprioception for integrated motor action feedback. *Neuron* **109**, 1188–1201.e7 (2021).
Discovers 'inner proprioceptors' (Piezo2$^+$ neurons) embedded in the spinal cord that detect body curvature; these neurons provide direct feedback to the CPG to regulate swimming dynamics, acting as an intrinsic system for monitoring motor output.
176. Jalalvand, E., Robertson, B., Tostivint, H., Wallén, P. & Grillner, S. The spinal cord has an intrinsic system for the control of pH. *Curr. Biol.* **26**, 1346–1351 (2016).
177. Jalalvand, E., Robertson, B., Wallén, P. & Grillner, S. Ciliated neurons lining the central canal sense both fluid movement and pH through ASIC3. *Nat. Commun.* **7**, 10002 (2016).
178. Wyart, C., Carbo-Tano, M., Cantaut-Belarif, Y., Orts-Del'Immagine, A. & Böhm, U. L. Cerebrospinal fluid-contacting neurons: multimodal cells with diverse roles in the CNS. *Nat. Rev. Neurosci.* **24**, 540–556 (2023).
179. Di Prisco, G. V., Walle, P. & Grillner, S. Synaptic effects of intraspinal stretch receptor neurons mediating movement-related feedback during locomotion. *Brain Res.* **530**, 161–166 (1990).
180. Grillner, S. & Wallén, P. On peripheral control mechanisms acting on the central pattern generators for swimming in the dogfish. *J. Exp. Biol.* **98**, 1–22 (1982).
181. Calabrese, R. L. & Marder, E. Degenerate neuronal and circuit mechanisms important for generating rhythmic motor patterns. *Physiol. Rev.* **105**, 95–135 (2025).
182. Katz, P. S. Evolution of central pattern generators and rhythmic behaviours. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* **371**, 20150057 (2016).
183. Kristan, W. B. Jr., Calabrese, R. L. & Friesen, W. O. Neuronal control of leech behavior. *Prog. Neurobiol.* **76**, 279–327 (2005).
184. Nusbaum, M. P., Blitz, D. M. & Marder, E. Functional consequences of neuropeptide and small-molecule co-transmission. *Nat. Rev. Neurosci.* **18**, 389–403 (2017).
185. Drapeau, P. et al. Development of the locomotor network in zebrafish. *Prog. Neurobiol.* **68**, 85–111 (2002).
186. Lacquaniti, F., Ivanenko, Y. P. & Zago, M. Development of human locomotion. *Curr. Opin. Neurobiol.* **22**, 822–828 (2012).
187. Landmesser, L. T. General principles of spinal motor circuit development: early contributions from research on avian embryos. *Int. J. Dev. Biol.* **62**, 235–243 (2018).
188. Vinay, L. et al. Perinatal development of the motor systems involved in postural control. *Neural Plast.* **12**, 131–139 (2005).
189. Combes, D., Merrywest, S., Simmers, J. & Sillar, K. T. Developmental segregation of spinal networks driving axial-and hindlimb-based locomotion in metamorphosing *Xenopus laevis*. *J. Physiol.* **559**, 17–24 (2004).
190. Chevallier, S., Jan Ijspeert, A., Ryczko, D., Nagy, F. & Cabelguen, J. M. Organisation of the spinal central pattern generators for locomotion in the salamander: biology and modelling. *Brain Res. Rev.* **57**, 147–161 (2008).
191. Ijspeert, A. J., Crespi, A., Ryczko, D. & Cabelguen, J. M. From swimming to walking with a salamander robot driven by a spinal cord model. *Science* **315**, 1416–1420 (2007).
192. Thandiackal, R. et al. Emergence of robust self-organized undulatory swimming based on local hydrodynamic force sensing. *Sci. Robot.* **6**, eabf6354 (2021).
193. Bagnall, M. W. & McLean, D. L. Modular organization of axial microcircuits in zebrafish. *Science* **343**, 197–200 (2014).
194. Menelaou, E. & McLean, D. L. Hierarchical control of locomotion by distinct types of spinal V2a interneurons in zebrafish. *Nat. Commun.* **10**, 4197 (2019).
195. Guan, N. N. et al. A specialized spinal circuit for command amplification and directionality during escape behavior. *Proc. Natl Acad. Sci. USA* **118**, e2106785118 (2021).

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